

Extreme-part conditions in integer partitions

(OEIS A350879, A237753, A237751, A118096 and A117086)

Part I: fixed slope partition extremes and inverse asymptotics

Part II: exact ratio two partitions at fixed asymptotic order

Part III: deficit asymptotics for partitions with divisible extreme parts

Three research reports of one external research session
(bundle Reports 216, 217 and 218), bound into one ProveIt report

All Parts: 4 October 2026 merged: 7 October 2026

Abstract

Three manuscripts on partitions λ of n constrained by a relation between their extreme parts, bound into one report. Part I fixes integers $k \geq 2$, $b \geq 0$ and counts partitions with largest part M and length L satisfying $M = kL + b$ or $M \geq kL + b$ (by conjugation, the columns of A350879; A237753 and A237751 at $k = 2$): exact finite-product sieves, expansions to every fixed order with leading coefficient $k!$, explicit corrections, and a Lambert W_{-1} inverse with a shrinking two-ceiling bracket. Part II counts partitions whose largest part is exactly twice the smallest (A118096): a complete-circle estimate, a discrete two-variable saddle and an exact finite coefficient rule give every fixed order on the scale $e^{\pi\sqrt{2n/15}}$, four explicit corrections, rounding-aware inverses, and eventual strict increase and log-concavity. Part III counts partitions whose largest part is divisible by the smallest (A117086) and proves that the deficit $D(n) = p(n) - a(n)$ satisfies $D(n) \sim \frac{\pi}{2\sqrt{6n}}p(n)$ with an expansion to every fixed order, fixed-minimum residue equidistribution, a stretched-exponential comparison with a reciprocal-minimum weighted count, a rational certificate at $n = 1000$, and inverse brackets. Parts I and III share the finite-shift Rademacher transfer; Part II's generating function is the $L = 2$ summand of Part III's. The write records that A350879's alternative elementary form, with $\exp(\sqrt{2\pi n/3})$, is false (the rate is $\exp(\pi\sqrt{2n/3})$), proves that A117086 increases strictly for every n , quotes the five OEIS entries, and compares the inverses with the repository's transseries volume. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit d451ef3d8 (batch 111).

Part	Bundle report	Delivered title	Labels
I	216 (base)	Fixed slope partition extremes and inverse asymptotics (4 October 2026)	<code>xpc:s1:</code>
II	217	Exact ratio two partitions at fixed asymptotic order (4 October 2026; source edition “v2”)	<code>xpc:r2:</code>
III	218	Deficit asymptotics for partitions with divisible extreme parts (4 October 2026)	<code>xpc:dv:</code>

What binds the Parts. All three count partitions through a condition coupling the extreme parts: $M = kL + b$ or $M \geq kL + b$ with M the largest part and L the length (Part I; by conjugation, M and L trade places, so these are also conditions on the smallest-part multiplicity), $\max = 2 \min$ (Part II) and $\min \mid \max$ (Part III). Part II’s generating function $\sum_k q^{3k} / \prod_{j=k}^{2k} (1 - q^j)$ is the $L = 2$ summand of (104): the partitions of Part II are exactly those of A117086 whose ratio $\max / \min$ equals 2. Parts I and III share their analytic machinery: a finite sieve or polynomial reduction whose discarded part is bounded on the entire Cauchy circle, and a finite-shift transfer from the Rademacher series with the same terminating polynomials Q_ℓ . Part II uses a different method (a discrete two-variable saddle at the golden-ratio point and Bessel identification) and a different scale.

	Part I (216)	Part II (217)	Part III (218)
Condition	$M = kL + b$, $M \geq kL + b$ ($k \geq 2$, $b \geq 0$ fixed)	$\max = 2 \min$	$\min \mid \max$ (the deficit $D = p - a$)
OEIS	A350879 columns; A237753, A237751	A118096	A117086
Scale	$k! B(n)t^k$ and $k! B(n)t^{k-1}$, $t \sim \pi / \sqrt{6n}$	$Cn^{-1/2} e^{\pi \sqrt{2n/15}}$	$B(n)t/2$
Orders	every fixed order; four coefficients for A237753, A237751	every fixed order; $c_1, \dots, c_4$	every fixed order; $c_0, \dots, c_5$
Method	finite sieve, circle bound, Rademacher shift transfer	complete-circle bound, discrete saddle, Bessel identification	polynomial reduction, first-harmonic gap, shift transfer
Inverse	W_{-1} core, triangular reversion, two ceilings	log-log polynomials P_j , two ceilings	W_{-1} core, triangular reversion, two ceilings; exact two-index shortcut
Also	fixed-offset calculus	strict log-concavity	residue equidistribution; weighted count; rational certificate

Arrangement and duplication. The base of the merge is Report 216: its `Report216.tex` was staged as this `article.tex`; it is the most general (a two-parameter family), it arrived first of the three, and it carries the batch’s one recorded OEIS correction. Report 217 is Part II and Report 218 Part III. No Part cites another. Part III re-derives the partition input that Part I proves: the one-term Rademacher gap, the terminating polynomials Q_ℓ and the finite-shift transfer (Section 27 against Lemmas 4.1 and 4.2 and Proposition 4.3), and the same triangular inverse reversion (Section 30 against Proposition 6.1). These are kept, with dated notes naming Part I’s

counterparts, because Part III's later statements cite them by their own labels; the proofs are the same argument.

Numbering. Sections are numbered continuously, statements within sections and equations globally, as delivered. Part I keeps all its numbers, including its Tables 1–2 and its Appendix A. Report 217's Section k is Section $k + 10$ here and its equation (k) is equation ($k + 65$); Report 218's Section k is Section $k + 21$, its equation (k) is equation ($k + 103$), and its appendix keeps the letter A. Statements move with their sections. Part labels carry `xpc:s1:`, `xpc:r2:` and `xpc:dv:`; the labels added by the write carry the Part prefix, or `xpc:` for this Guide. The bibliography is merged at the end: the OEIS entries of Parts II and III (both delivered as `oeis`) are `oeisA118096` and `oeisA117086`; Part II's `d1mf` (Bessel sections) is `d1mfBessel`; Parts I and III's `d1mf` (Section 23.18) and `johansson` are printed once; Part III's `nr` is Part I's `ngorhoades` (the same paper).

[write] **Provenance, added 7 October 2026, batch 111.** This report was built from three manuscripts, all dated 4 October 2026, arrived in `60f54ea06` and placed in `d451ef3d8`. *Part I:* Report 216, archive `Report216-reproducibility.zip` (567,001 bytes, SHA-256 `ca858cb05b9b...ea13edb9ec180c7`, 20 files in a wrapper directory `Report216/`); text `Report216.tex` (738 lines, 20 pages as delivered). *Part II:* Report 217, archive `Report217-source-v2.zip` (409,022 bytes, SHA-256 `96812623e011...abcc67943772177`, 21 files in `Report217/`); text `article.tex` (594 lines, 14 pages); the archive name carries the edition mark “v2”, the only such mark in the bundle; no other edition was delivered. *Part III:* Report 218, archive `Report218-reproducibility.zip` (544,706 bytes, SHA-256 `f688cd814dde...7e4377cbbb2b09f`, 17 files in `Report218/`); text `report218.tex` (976 lines, 22 pages). Report 216's title page has an empty author line and an empty PDF author field; Reports 217 and 218 read “Report 217” and “Report 218”. None names a person, tool or addressee; none carries a “prepared for private review” line, an e-mail address or personal data, and none records a ProveIt commit. Each Part's programs are shipped under `code/` and its data, generated tables and recorded outputs under `data/`, with the prefixes `216-slope-`, `217-ratio2-` and `218-divisible-`; Report 216's `SOURCES.txt`, `SOURCE_FILES.txt` and code `README`, Report 217's `SOURCE_NOTES.md` and code `README`, and Report 218's code `README` are shipped at the top level under the same prefixes; Report 217's `MANIFEST.json`, which also records its claim and source-comparison fields, is shipped as `data/217-ratio2-MANIFEST.json`. Report 216's delivered `README.txt` was staged as the report `README` and is replaced by it. Not shipped, and retrievable from the arrival commit: the three delivered PDFs, the texts of Reports 217 and 218 (they are Parts II and III of this file), their delivered `READMEs`, and the pure checksum manifests (Report 216 `MANIFEST.json`, 19 entries; Report 217 `generated/SHA256SUMS.json`, 5; Report 218 `MANIFEST.json` and `SHA256SUMS`, 15 and 16), all verified at the write, as was Report 217's `MANIFEST.json` (20 entries). Report 216's two generated tables and Report 218's four generated tables, which the delivered texts read with `\input`, are printed inline here (the shipped files under `data/` are byte-identical to the delivered ones). Where the merge had to choose: the base and the order; the citation keys above; keeping Part III's re-derivations; the per-Part paragraph and list spacing of the deliveries (Part I sets no paragraph indent, Part II tight lists, Part III looser lists and allowed display breaks), restored at each Part header. The write prefixed the 92, 54 and 90 delivered labels (Report 216's 92 include the two table labels of its generated tables) and updated the 83, 57 and 86 references to them (66, 36 and 60 `\eqref`; 17, 21 and 26 `\ref`); it labelled three unlabelled sections of Report 217 and two of Report 218, and added this Guide, the Part headers, the remarks marked [write] (Remarks 1.2, 1.3, 6.5, 11.2, 17.2, 22.2, 30.3) and the dated notes. The added remarks are the last statements of their sections and the added displays are unnumbered, so every delivered number is unchanged up to the stated shifts (label tables of builds of the three delivered texts and of this one compared).

The sources, as the write read them (7 October 2026). OEIS A350879, A237753, A237751, A118096 and A117086 with their b-files, in the internal format (Remarks 1.2, 11.2 and 22.2); the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (`p0:def:model`, `p0:def:core`, `p0:thm:lambert-core`, `p0:thm:lambert-centered`, `p0:thm:flattening`, `p0:thm:backward-error`, `p0:def:three-inverses`, `p0:thm:staircase`). The write tried to open Pittel's 2007 paper at its publisher (refused, HTTP 403); like the source, it did not read it. Not read by the write: the rest of the cited literature; the Parts' accounts of what they inspected are reported, not confirmed.

What was checked at intake. The batch-111 dossier (7 October 2026) read the three manuscripts, found no error in them, confirmed the A350879 discrepancy against the live entry, and reran the three delivered suites on copies: their outputs reproduce, the only failures being Windows artifacts (line-ending bytes in hashed outputs, a recorded Python version string, a console code page), none mathematical. *What the write checked.* Every proof line by line, with no error found. With its own code (Python 3.14.4, SymPy 1.14.0, mpmath 1.3.0) it recomputed, for Part I, $E_{2,0}$ and $T_{2,1}$ by the sieve to $n = 3000$ (all A237753 and A237751 b-file terms in range agree, as do brute-force counts for $n \leq 40$ and all columns $k \geq 2$ of A350879's 50-row b-file), the shipped selected counts, the radial series (45)–(47), the coefficients (46) and (48), the general first coefficients (15)–(16) for $2 \leq k \leq 6$, $0 \leq b \leq 3$, the inverse coefficients (55)–(57) by solving (54), and (62); for Part II, A118096 to $n = 3000$ from (66) (b-file and brute force agree), the saddle data $H_2, H_3, H_4, \ell'(x_0), \ell''(x_0), e_1, e_3, h(x_0) = A, \lambda$, the four coefficients (70) from the finite rule (88) (see the note at the end of Section 15), $d_1, \dots, d_4$ both from (71) and from (69), P_1 and P_2 , and the diagnostics of Section 19; for Part III, the counts against A117086's b-file and brute force, $D(17), D(18)$ and $D(1000)$, $g_1, \dots, g_8$ from both (133) and (134), $c_0, \dots, c_5, \rho_1, \rho_2, \rho_3, d_1^{(D)}, d_2^{(D)}, d_3^{(D)}$, the general $v_1, v_2, v_3, v_{1,2}^{(D)}, v_{1,2}^{(a)}$, the centres (164), (165) and (167), $s_3(1000), u_3(1000)$ and the two bounds of (171) from the printed E (the certificate's E itself was not recomputed), and the rational inequalities of Section 29. Finite and floating computations are observations, not certificates.

Relation to the repository. Before batch 111 no file of the repository named A350879, A237753, A237751, A118096 or A117086. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The nearest reports are in the same directory `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`: `a239950-maximal-schreier-supports` (batch 111: least part equal to the number of distinct sizes, another extreme-part condition, on a different exponential scale) and `a239964-sizes-equal-max-multiplicity` (batch 111). They share no result with this report, so no reciprocal note is proposed. The inverses are compared with the transseries volume in Remarks 6.5, 17.2 and 30.3.

[write] **What is not claimed, collected from the three Parts.** *Part I.* Fixed slope, fixed offset and fixed order only: k, b and R may not grow with n , no convergence or numerical error bound for the formal series, no effective onset, no $k = 1$ (rank) conclusion, no growing-offset sum; the inverse constants are not effective and the bracket is not a certified finite- y enclosure; no historical priority: the leading equality law is Kotěšovec's, partition-product transfer is not new, the comparison with Szekeres, Jiang–Wang, Melczer–Panova–Pemantle, Richmond, Hwang, Ngo–Rhoades, Dyson and Berkovich–Garvan is bounded; the algebraic checks are not clean-room and do not form a separate proof; diagnostics are not certified; same-toolchain byte stability only. *Part II.* No novelty claim and no solution of a known open problem: the leading equivalent is Kotěšovec's (2025), the radial all-order machinery is McIntosh's and Zhou's, and the comparison with Pittel (2007) is unresolved (abstract only; “not evidence of absence of overlapping results”); no uniformity in the order, no explicit threshold, no effective last exception for increase or log-concavity; the inversion and ceiling mechanisms are standard; diagnostics are not certified; $H(q)$ is not identified with a mock theta function. *Part III.* No historical priority, no uniform growing-order expansion, no convergence; the twisted product, all-order transfer and arbitrary-weight tail frameworks are prior (Bridges–Franke–Garnowski, Ngo–Rhoades, Cesana–Craig–Males, Andrews–Freitas, the last not read in the original); the stretched-exponential bound is an upper bound, not an asymptotic, and proves no $\exp(-c\sqrt{n})$ rate, and its optimality concerns only the two-term majorant; no global monotonicity of D ; the inverse constants are ineffective and no blind rounding is licensed; no quantum-modularity claim; the A117086 revision history was not inspected. *The write adds:* its checks are floating, finite or symbolic computations; the three transseries remarks claim no novelty for any inversion; the merge claims no result connecting the Parts beyond the shared summand and machinery.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check re-derived every [write] statement of this Guide and of the three Parts with its own code, after fetching the five OEIS entries and their b-files again. *Confirmed, hardest first:* the refutation of Remark 1.3 (the live A350879 comment is quoted verbatim; $2^{(k+4)/2}3^{(k+1)/2} = 4\sqrt{3}6^{k/2}$; with $A237753(n)$ from the b-file, the quotient of the posted first form by the true equivalent equals $\exp(-(\pi - \sqrt{\pi})\sqrt{2n/3})$, and $A237753(3000)$ is $0.7274\dots$ times the true equivalent and $2.84\dots \cdot 10^{26}$ times the posted form; A237753's own formula line agrees with the true one at $k = 2$); the proof that A117086 increases strictly for every $n \geq 0$ (Remark 30.3(a)); the counts, by brute force for $n \leq 45$ (all four statistics and every A350879 entry), by Manyama's column generating

function to $n = 1000$, by the sieves (19) and (20) with the check's own code to $n = 3000$, against all b-file terms in range; the radial series and the coefficients of (46), (48), the general c_1 for $2 \leq k \leq 5$, $0 \leq b \leq 3$; $d_1, \dots, d_4$ of Part II by both routes, the two relative errors at $n = 10000$ and the inverse offset $2.709 \dots \cdot 10^{-5}$; $g_1, \dots, g_8$ and $c_0, \dots, c_5$ of Part III; the monotonicity and log-concavity observations; the label tables (Part I unchanged; Part II shifted by 10 sections and 65 equations, Part III by 21 and 103, appendix letters kept; 92, 54, 90 delivered and 16 added labels, 252 in all; 83, 57, 86 references); the merged bibliography; the transseries classifications; and the three reruns from the shipped files (identical up to line endings and, for Report 216, the recorded Python version). *Corrections and precisions.* (i) In the provenance note above, Reports 217 and 218 read “Report 217” and “Report 218” on their title pages only: their PDF author fields are empty, like Report 216's. (ii) In item (a) of each of Remarks 6.5, 17.2 and 30.3, `p0:thm:staircase(1)` is stated for targets at least the sequence value at the onset n_1 , in addition to exceeding every earlier value; both hold for all sufficiently large targets, which is how those items are to be read. No other claim of the write or of the sources was found wrong or unproved.

Reading conventions (letters the Parts use differently, with the tempting false readings; no symbol of any source was renamed; symbols of the transseries volume carry the subscript “vol” in the three transseries remarks).

Symbol	Part I	Part II	Part III
A	$\pi^2/6$	$\pi^2/30$ (and a the curvature $3\phi - 4$)	$A = \pi^2/6$
B	$B(n)$, the partition scale	$2\sqrt{A}$	$\mathcal{B}(n)$, the partition scale
C	C_M , an inverse constant	$1/\sqrt{5 + \sqrt{5}}$	$C_D, C_a; C_0, C_1$ explicit constants
D	D_j , sieve exponents; the inverse constant D	$D = 2A - ax_0^2$	$D(n) = p(n) - a(n)$, the deficit
$a(n), X$	$X \in \{E, T\}$	$a(n) = A118096; X_J(Y)$ the inverse	$a(n) = A117086; X \in \{D, a\}$
t, s	$t = \sqrt{A/(n - \frac{1}{24})}; s = s_X$ the valuation	$t = \sqrt{A/n}; s = \sqrt{t}$, but $s = B\sqrt{n}$ in Section 17	$t = \sqrt{A/N}, N = n - \frac{1}{24}$
c_m, c_j	relative coefficients in t	radial coefficients in t (and d_r in $n^{-1/2}$)	relative coefficients in t
d, d_j	$d = s + 2$	d_r	d_j , coefficients in w^{-1}
G, g_ℓ	$G_{k,b}^X, g_\ell$	$g(x)$ the amplitude	$G(z), g_\ell; g(x)$ in Section 28
H	$H(q)$ a sieve polynomial; $H(v)$ the forward series	$H_m = h^{(m)}(x_0); H(q)$ the weak-class series; H_J	$H_k(z, q); H(q)$ the weighted series; H an exponential degree
Q	$Q_\ell(t)$	$Q(y, v)$ the quadratic form	$Q_\ell(t)$ (the same as Part I's)
T	$T_{k,b}$, the tail count	$T_k(q)$ summands; $T = \log(Y/(CB))$	$T_K(q)$, the minimum tail
L	$L(\lambda)$, the length	L a block length; $L_t; L_J(x)$	L in (104); $L = \log(1/t)$ in Section 29
w, w_0	$w = 2\sqrt{A(n - \frac{1}{24})}$, w_0 the Lambert root	$w = \log T$	$w = 2\sqrt{A(n - \frac{1}{24})}$, w_0
ν, N	$\nu_X(y)$ thresholds; $N = n - \frac{1}{24}$	$N(Y)$ the threshold	$\nu_X(y); N = n - \frac{1}{24}$
u_j, v_j, Δ	u_j , the inverse coefficients	—	v_j , the same coefficients for data $d_j = (2A)^j c_j$
k, b, M	slope, offset; M the largest part, also an inverse order	k the smallest part; M an order	k the minimum; b a residue; M the maximum, an order

Symbol	Part I	Part II	Part III
$X_{\text{vol}}, a_{\text{vol}}, b_{\text{vol}}, L_{\text{vol}}$	the transseries volume's core variable and data, never the Parts' X, a, b, L		

Part I

Fixed slope partition extremes and inverse asymptotics

Bundle Report 216, archive `Report216-reproducibility.zip`, delivered as `Report216.tex` (20 pages), the base of the merge. Delivered title block “Fixed slope partition extremes and inverse asymptotics / Report216 / 4 October 2026” (empty author line). Printed as delivered, with `[write]` remarks and notes; section, statement, equation and table numbers are those of the delivery, its labels carry `xpc:sl:`, and its two generated tables are printed inline. This Part is self-contained.

Abstract of this Part (as delivered). Fix integers $k \geq 2$ and $b \geq 0$. We count partitions of n whose largest part M and number of parts L satisfy $M = kL + b$ or $M \geq kL + b$. Exact finite-product sieves give, for every fixed order, explicit expansions in powers of $(n - 1/24)^{-1/2}$ relative to the principal partition scale. The leading coefficients are $k!$, with powers k and $k - 1$, respectively. The proof controls the discarded sieve terms on an entire Cauchy circle, including its $t^{-3/2}$ loss, then transfers each retained polynomial by a finite-shift form of the Rademacher formula. We give rational coefficient generators, the first four coefficients for A237753 and A237751, fixed-offset companions, and a triangular Lambert W_{-1} inverse with shrinking two-ceiling uncertainty for the first index whose count is at least a target. Kotěšovec’s posted leading equality equivalent is credited; classical joint-extreme and general transfer results are compared in their actual parameter regimes. The claims concern fixed slope, fixed offset, and fixed order. Numerical diagnostics are not certified error bounds, and no historical priority, effective onset, or growing-order assertion is made.

1 The statistics and the result

For a nonempty integer partition λ , let $M(\lambda)$ be its largest part and $L(\lambda)$ its number of positive parts. Throughout this article $k \geq 2$ and $b \geq 0$ are fixed integers. Define

$$E_{k,b}(n) = \#\{\lambda \vdash n : M(\lambda) = kL(\lambda) + b\}, \quad (1)$$

$$T_{k,b}(n) = \#\{\lambda \vdash n : M(\lambda) \geq kL(\lambda) + b\}. \quad (2)$$

The empty partition is excluded, so both counts are zero at $n = 0$. Let $p(n)$ denote the unrestricted partition number, with $p(0) = 1$ and $p(n) = 0$ for negative integers n .

Conjugation interchanges M and L . It therefore identifies $E_{k,0}$ with column k of OEIS A350879, whose displayed convention is $kM = L$. In particular,

$$E_{2,0}(n) = A237753(n), \quad T_{2,1}(n) = A237751(n). \quad (3)$$

For the second identity, the OEIS condition $2M < L$ conjugates to $M > 2L$, which is $M \geq 2L + 1$ on integers. The offset 1 is essential for the lower-order coefficients.

The leading equality equivalent

$$E_{k,0}(n) \sim k! \left(\frac{\pi}{\sqrt{6n}} \right)^k p(n) \quad (4)$$

is already attributed to Václav Kotěšovec, 17 October 2024, in A350879 and its $k = 2$ specialization [1, 2]. The equality generating function at $b = 0$ is also recorded there. Our purpose is a fully justified,

explicit fixed-order expansion and inversion for both statistics with fixed offsets. We do not claim that the leading scale or the general idea of partition-product transfer is new. Section 8 explains the relationship with the relevant primary literature.

Set

$$A = \frac{\pi^2}{6}, \quad N = n - \frac{1}{24}, \quad t = \sqrt{\frac{A}{N}}, \quad B(n) = \frac{\exp(2\sqrt{AN})}{4\sqrt{3}N}. \quad (5)$$

The parameter t is positive for integers $n \geq 1$ and tends to zero. Write $X = E$ or $X = T$ for the kind of statistic and define

$$\epsilon_E = 1, \quad \epsilon_T = 0, \quad s_E = k, \quad s_T = k - 1. \quad (6)$$

We sometimes write simply $s = s_X$ and $X_{k,b}$.

For $j \geq 1$ let

$$D_j = D_j(k, b) = \frac{(2k+1)j^2 + (2b-1)j}{2}. \quad (7)$$

This is an integer: the numerator is even because $j^2 - j$ is even. Define formal power series

$$G_{k,b}^E(z) = \sum_{j \geq 1} (-1)^{j-1} e^{-D_j z} \prod_{r=j}^{kj} (1 - e^{-rz}), \quad (8)$$

$$G_{k,b}^T(z) = \sum_{j \geq 1} (-1)^{j-1} e^{-D_j z} \prod_{r=j+1}^{kj} (1 - e^{-rz}). \quad (9)$$

The j th summand has valuation $(k-1)j + \epsilon_X$. Thus every coefficient uses finitely many terms; no convergence of the infinite series near $z = 0$ is asserted or needed. Put $g_\ell = [z^\ell] G_{k,b}^X(z)$.

Define terminating polynomials

$$Q_\ell(t) = \sum_{h=0}^{\ell+1} \frac{(-1)^h (\ell+h+1)!}{(\ell+1-h)! h!} \left(\frac{t}{4A} \right)^h \quad (\ell \geq 0), \quad (10)$$

and define $c_m = c_m(k, b, X)$ by the formal identity

$$\sum_{m \geq 0} c_m t^m = \frac{1}{k! t^s} \sum_{\ell \geq s} g_\ell t^\ell Q_\ell(t). \quad (11)$$

Each c_m is a polynomial in A^{-1} with rational coefficients. This is also a finite algorithm: to compute $c_0, \dots, c_R$, only $g_s, \dots, g_{s+R}$ are required.

Theorem 1.1 (Every fixed order). *For every fixed nonnegative integer R ,*

$$X_{k,b}(n) = k! B(n) t^s \left(\sum_{m=0}^R c_m(k, b, X) t^m + O_{k,b,R}(t^{R+1}) \right). \quad (12)$$

Here $c_0 = 1$, and the coefficients are the explicit rational polynomials defined by (8)–(11). In particular,

$$E_{k,b}(n) \sim k! \left(\frac{\pi}{\sqrt{6n}} \right)^k p(n), \quad (13)$$

$$T_{k,b}(n) \sim k! \left(\frac{\pi}{\sqrt{6n}} \right)^{k-1} p(n). \quad (14)$$

These are fixed-depth asymptotic statements. The constants and the onset may depend on k, b, R . They do not permit k, b , or R to grow with n , and do not imply convergence or a numerical error bound for the full formal series.

The first relative coefficients are

$$c_1(k, b, E) = -\frac{k(k+5)}{4} - b - 12\mathbf{1}_{\{k=2\}} - \frac{(k+1)(k+2)}{4A}, \quad (15)$$

$$c_1(k, b, T) = -\frac{k(k+5) - 2}{4} - b - 6\mathbf{1}_{\{k=2\}} - \frac{k(k+1)}{4A}. \quad (16)$$

In particular the leading tail companion for A237751 is

$$A237751(n) \sim \frac{2\pi}{\sqrt{6n}} p(n). \quad (17)$$

This is a consequence of the theorem, not a claim of historical firstness.

A literal discrepancy in the posted equivalent. At the source check on 4 October 2026, A350879 printed an alternative elementary exponential expression with $\exp(\sqrt{2\pi n/3})$, while its partition-normalized expression is (4). The correct rate equivalent to (4) is $\exp(\pi\sqrt{2n/3})$. Indeed the usual equivalent for $p(n)$ gives

$$E_{k,0}(n) \sim \frac{k!\pi^k}{4\sqrt{3}6^{k/2}n^{1+k/2}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right). \quad (18)$$

The discrepancy is a missing factor of π under the square root in that alternative expression. It does not invalidate the already posted $p(n)$ -normalized leading law. No OEIS edit is part of this work.

Remark 1.2 ([write] The OEIS entries, quoted, 7 October 2026). Read on 7 October 2026 in the internal format.

- *A350879*, revision #51 (18 October 2024), author Seiichi Manyama (21 January 2022): “Triangle $T(n,k)$, $n \geq 1$, $1 \leq k \leq n$, read by rows, where $T(n,k)$ is the number of partitions of n such that $k^*(\text{greatest part}) = (\text{number of parts})$.”; a comment gives the conjugate reading “(greatest part) = $k^*(\text{number of parts})$ ”; the column generating function of Manyama; columns $k = 1, \dots, 7$ are A047993, A237753, A237756, A348163, A348164, A377107, A377108; b-file by Andrew Howroyd (rows 1–50). Kotěšovec’s comment of 17 October 2024 reads: “Column $k > 1$ is asymptotic to $k! \cdot \pi^k \cdot \exp(\sqrt{2\pi n/3}) / (2^{((k+4)/2)} \cdot 3^{((k+1)/2)} \cdot n^{((k+2)/2)})$. Equivalently, for fixed $k > 1$, $T(n,k) \sim k! \cdot \pi^k \cdot A000041(n) / (6^{(k/2)} \cdot n^{(k/2)})$.” The first form is false; see Remark 1.3.
- *A237753*, revision #32 (11 July 2026), author Clark Kimberling (13 February 2014): “Number of partitions of n such that $2^*(\text{greatest part}) = (\text{number of parts})$.”; Manyama’s generating function; Kotěšovec’s “ $a(n) \sim \pi^2 \cdot \exp(\pi\sqrt{2n/3}) / (4 \cdot 3^{(3/2)} \cdot n^2)$ ” (17 October 2024), which is (18) at $k = 2$ and is correct; b-file to $n = 10000$ (Kotěšovec, terms to 1000 from Manyama).
- *A237751*, revision #11 (24 January 2022), author Clark Kimberling (13 February 2014): “Number of partitions of n such that $2^*(\text{greatest part}) < (\text{number of parts})$.”; it adds the conjugate reading “(greatest part) $> 2^*(\text{number of parts})$ ” and “ $a(n) = A000041(n) - A237755(n)$.”; b-file by Manyama to $n = 1000$. It states no asymptotic formula; (17) is not recorded there.

The write computed $E_{2,0}$ and $T_{2,1}$ from (19) and (20) with its own code for $n \leq 3000$: every term of the A237753 and A237751 b-files in that range agrees, as do brute-force counts for $n \leq 40$; and $E_{k,0}(n)$ agrees with every entry of columns $k \geq 2$ of A350879's b-file. Nothing was submitted to the OEIS.

Remark 1.3 ([write] A350879's alternative form is false, 7 October 2026). *Claim.* For every fixed $k \geq 2$ the first form of Kotěšovec's comment in A350879 (Remark 1.2),

$$T(n, k) \sim \frac{k! \pi^k \exp(\sqrt{2\pi n/3})}{2^{(k+4)/2} 3^{(k+1)/2} n^{(k+2)/2}},$$

is false, while its second form, the $p(n)$ -normalized law (4), is true. *Proof.* By conjugation $T(n, k) = E_{k,0}(n)$, and (18) (Theorem 1.1 with the Hardy–Ramanujan equivalent of $p(n)$) gives $T(n, k) \sim k! \pi^k \exp(\pi \sqrt{2n/3}) / (4\sqrt{3} 6^{k/2} n^{1+k/2})$. Since $2^{(k+4)/2} 3^{(k+1)/2} = 4\sqrt{3} 6^{k/2}$, the quotient of the posted first form by this equivalent is $\exp(-(\pi - \sqrt{\pi})\sqrt{2n/3}) \rightarrow 0$, so the first form is not asymptotic to $T(n, k)$. Only the exponential rate is wrong: $\sqrt{2\pi n/3}$ lacks a factor π under the root, as the source says in Section 1; the constant and the power of n are right. *Numerically* (not part of the proof), A237753(3000) is 0.7274... times the correct equivalent and $2.84 \dots \cdot 10^{26}$ times the posted first form. No OEIS edit was made.

2 Exact finite product sieves

For $|q| < 1$, write

$$(a; q)_m = \prod_{r=0}^{m-1} (1 - aq^r), \quad (q; q)_m = \prod_{r=1}^m (1 - q^r), \quad P(q) = \frac{1}{(q; q)_\infty} = \sum_{n \geq 0} p(n) q^n.$$

Empty finite products equal one.

Proposition 2.1 (Exact identities). *As formal power series and as analytic functions for $|q| < 1$,*

$$F_{k,b}^E(q) := \sum_{n \geq 0} E_{k,b}(n) q^n = P(q) \sum_{j \geq 1} (-1)^{j-1} q^{D_j} \prod_{r=j}^{kj} (1 - q^r), \quad (19)$$

$$F_{k,b}^T(q) := \sum_{n \geq 0} T_{k,b}(n) q^n = P(q) \sum_{j \geq 1} (-1)^{j-1} q^{D_j} \prod_{r=j+1}^{kj} (1 - q^r). \quad (20)$$

For every fixed coefficient q^n , only indices with $D_j \leq n$ contribute.

Proof. For a fixed length $l \geq 1$, subtracting one from every part gives the unrestricted generating function $q^l / (q; q)_l$. If $M \leq kl + b - 1$, the remaining diagram fits a rectangle of height l and width $kl + b - 2$. Its generating function is

$$q^l \frac{(q^{kl+b-1}; q)_l}{(q; q)_l}.$$

The fixed-length tail is therefore

$$F_{k,b}^T(q) = \sum_{l \geq 1} \frac{q^l}{(q; q)_l} (1 - (q^{kl+b-1}; q)_l). \quad (21)$$

Using the finite q -binomial identity in its explicit form

$$(a; q)_l = \sum_{j=0}^l \frac{(q; q)_l}{(q; q)_j (q; q)_{l-j}} (-a)^j q^{j(j-1)/2}, \quad (22)$$

the j th selected term in (21) is

$$\frac{(-1)^{j-1} q^{l+(kl+b-1)j+j(j-1)/2}}{(q; q)_j (q; q)_{l-j}} \quad (1 \leq j \leq l).$$

Set $l = j + h$. The exponent becomes $D_j + (kj + 1)h$, and Euler's identity gives

$$\sum_{h \geq 0} \frac{q^{(kj+1)h}}{(q; q)_h} = \frac{1}{(q^{kj+1}; q)_\infty} = P(q)(q; q)_{kj}. \quad (23)$$

After division by $(q; q)_j$ the remaining factors are $j + 1, \dots, kj$, proving (20). Finally

$$E_{k,b}(n) = T_{k,b}(n) - T_{k,b+1}(n), \quad D_j(k, b+1) = D_j(k, b) + j, \quad (24)$$

so subtraction inserts the missing factor $1 - q^j$ and gives (19).

All manipulations are coefficientwise legitimate: the exponents are nonnegative, and only finitely many length and sieve indices can affect a fixed degree; in particular $D_j \rightarrow \infty$ quadratically. The absolute bounds in Section 3 justify the analytic interchanges near the unit circle. Any smaller closed disk is dominated by a larger such radius, so they also establish the analytic assertions throughout the open disk. $\square$

For the equality remainder it is advantageous to start from a different exact expression, instead of subtracting two separately bounded tails.

Lemma 2.2 (Equality rectangle). *One has*

$$F_{k,b}^E(q) = \sum_{m \geq 0} q^{(k+1)m+k+b} \frac{(q^{k(m+1)+b}; q)_m}{(q; q)_m}. \quad (25)$$

Expanding its numerator through elementary-symmetric degrees $i = 0, \dots, J - 1$ and summing over m gives precisely the terms $j = 1, \dots, J$ in (19).

Proof. Take $L = m + 1$ and $M = k(m + 1) + b$. Removing the first row and first column removes $M + L - 1 = (k + 1)m + k + b$ cells. What remains is an arbitrary diagram in a rectangle of height m and width $M - 1$. Its Gaussian-binomial generating function is

$$\frac{(q^M; q)_m}{(q; q)_m},$$

which proves (25). Selecting degree i in (22) and then putting $m = i + h$, the h coefficient in the exponent is $k(i + 1) + 1$. Applying (23) with $j = i + 1$ leaves $(q; q)_{kj} / (q; q)_{j-1}$, namely the factors $j, \dots, kj$, with exponent D_j . Thus the truncation indices agree exactly, including $i = 0 \leftrightarrow j = 1$. $\square$

3 A remainder bound on the entire circle

A radial expansion of $G_{k,b}^X(t)$ alone does not establish a coefficient asymptotic. We instead truncate before extracting coefficients and control the discarded terms at every angle on a full circle.

Let $r = e^{-t}$, $\delta = 1 - r$, and $|q| = r$, with $t > 0$ tending to zero. Then $\delta \sim t$. For complex numbers $x_1, \dots, x_m$, put $S = \sum_i |x_i|$. If a product $\prod_i (1 - x_i)$ is truncated before elementary-symmetric degree J , its remainder is bounded by

$$\sum_{i \geq J} e_i(|x_1|, \dots, |x_m|) \leq \sum_{i \geq J} \frac{S^i}{i!} \leq \frac{S^J e^S}{J!}. \quad (26)$$

This follows from $i!e_i \leq S^i$ and $(J+h)! \geq J!h!$. Also

$$\begin{aligned} \frac{1}{|(q; q)_m|} &\leq \frac{1}{(r; r)_m} = P(r)(r^{m+1}; r)_\infty \\ &\leq P(r) \exp\left(-\frac{r^{m+1}}{\delta}\right), \end{aligned} \quad (27)$$

using $\log(1-x) \leq -x$ for $0 < x < 1$.

Lemma 3.1 (Geometric mesh). *For fixed real numbers $a, c > 0$, and $u_m = r^{m+1}$,*

$$\sum_{m \geq 0} u_m^a \exp(-cu_m/\delta) = O_{a,c}(\delta^a/t). \quad (28)$$

Proof. The function $h(x) = \exp(-ax - ce^{-x}/\delta)$ is unimodal on $[0, \infty)$. Comparing each side of its maximum with the corresponding integral bounds its mesh sum by

$$\sum_{m \geq 0} h(t(m+1)) \leq t^{-1} \int_0^\infty h(x) dx + 2 \sup_{x \geq 0} h(x).$$

With $u = e^{-x}$,

$$\int_0^\infty h(x) dx = \int_0^1 u^{a-1} e^{-cu/\delta} du \leq \Gamma(a)(\delta/c)^a.$$

The maximum of $u^a e^{-cu/\delta}$ on $u > 0$ is $(a\delta/(ce))^a$. Since $t \rightarrow 0$, these bounds give (28). $\square$

Write $F_{k,b,J}^X$ for (19) or (20) with only $j = 1, \dots, J$ retained. For $J = 0$ the truncated expression is zero.

Proposition 3.2 (Global finite-depth bound). *For every fixed nonnegative integer J ,*

$$\sup_{|q|=r} |F_{k,b}^E(q) - F_{k,b,J}^E(q)| = O_{k,b,J}\left(P(r)t^{k+(k-1)J}\right), \quad (29)$$

$$\sup_{|q|=r} |F_{k,b}^T(q) - F_{k,b,J}^T(q)| = O_{k,b,J}\left(P(r)t^{(k-1)(J+1)}\right). \quad (30)$$

Proof. For equality, use (25) and set $u = r^{m+1}$. The sum of the numerator-factor moduli is

$$S_m = r^b u^k \frac{1 - r^m}{\delta} \leq \frac{u^k(1-u)}{\delta}.$$

Here $b \geq 0$ and $r^m = u/r \geq u$. Since $k \geq 2$,

$$u^{k-1}(1-u) \leq u(1-u) \leq \frac{1}{4} \quad (0 < u < 1). \quad (31)$$

Consequently $-u/\delta + S_m \leq -3u/(4\delta)$. The exterior power in (25) has modulus $r^{b-1}u^{k+1} \leq r^{-1}u^{k+1}$. Combining (26) and (27), the error after degrees $0, \dots, J-1$ is at most

$$\frac{P(r)r^{-1}}{J!\delta^J} \sum_{m \geq 0} u^{k+1+kJ} \exp\left(-\frac{3u}{4\delta}\right).$$

The mesh lemma makes this

$$O\left(P(r) \frac{\delta^{k+1+(k-1)J}}{t}\right) = O\left(P(r)t^{k+(k-1)J}\right).$$

Lemma 2.2 identifies the retained terms, proving (29).

For the tail, use (21) and put $u = r^{l+1}$. Now

$$S_l = r^{b-k-1}u^k \frac{1-u/r}{\delta} \leq r^{-k-1} \frac{u^k(1-u)}{\delta}.$$

For r sufficiently close to 1, depending only on the fixed k , we have $r^{-k-1} \leq 2$. Thus (31) gives

$$-u/\delta + S_l \leq -u/(2\delta).$$

The prefactor r^l equals $r^{-1}u$. Retaining degrees $1, \dots, J$ in $1 - \prod(1 - x_i)$ leaves degree at least $J+1$. Its error is bounded by a constant times

$$P(r)\delta^{-J-1} \sum_{l \geq 1} u^{1+k(J+1)} \exp(-u/(2\delta)).$$

The mesh estimate gives the power

$$\delta^{-J-1} \frac{\delta^{1+k(J+1)}}{t} = O\left(t^{(k-1)(J+1)}\right),$$

proving (30).

The same bounds, with $J = 0$ and the numerator absolute sum bounded by e^{S_m} or e^{S_l} , prove absolute summability of the expanded products. For the tail one may also use $e^{S_l} - 1 \leq S_l e^{S_l}$. This supplies the absolute interchange used in Proposition 2.1. Every inequality has been expressed in r , so the estimates hold on the entire circle, not only on its positive radius. $\square$

3.1 The Cauchy cost and the required depth

The eta inversion formula [14, 23.18.5–23.18.7] yields the exact identity

$$P(e^{-t}) = \sqrt{\frac{t}{2\pi}} \exp\left(\frac{A}{t} - \frac{t}{24}\right) P(e^{-4\pi^2/t}). \quad (32)$$

For clarity, substituting $\tau = it/(2\pi)$ into $\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau)$ and using $\eta(\tau) = e^{\pi i \tau/12} \prod_{r \geq 1} (1 - e^{2\pi i r \tau})$ gives precisely (32). The last factor equals $1 + O(e^{-4\pi^2/t})$ by its convergent power series at zero.

Choose t as in (5). Because $Nt = A/t$,

$$\frac{r^{-n}P(r)}{B(n)} = \frac{4\sqrt{3}A}{\sqrt{2\pi}}t^{-3/2} \left(1 + O(e^{-4\pi^2/t})\right). \quad (33)$$

Cauchy's coefficient inequality and Proposition 3.2 therefore give remainders

$$\begin{aligned} O\left(B(n)t^{k+(k-1)J-3/2}\right) & \text{ for equality,} \\ O\left(B(n)t^{(k-1)(J+1)-3/2}\right) & \text{ for the tail.} \end{aligned}$$

Relative to $B(n)t^s$, both powers are $(k-1)J - 3/2$. Thus the convenient integer choice

$$J \geq \left\lceil \frac{R+3}{k-1} \right\rceil \quad (34)$$

ensures a remainder $O(B(n)t^{s+R+1})$. In fact the left side of $(k-1)J \geq R + 5/2$ is integral, so (34) is the least depth guaranteed by this particular estimate. The loss $t^{-3/2}$ has been included; a merely radial valuation count would miss it.

For this J , the retained expression is $P(q)$ times a fixed polynomial in q . Its asymptotic analysis requires only finitely many shifted partition numbers, which we now control separately.

4 Finite shift transfer from the partition formula

We use the Rademacher formula in Johansson's exact U -series normalization [12, (1.4)–(1.8)], rather than using a leading Hardy–Ramanujan equivalent as if it were an all-order error estimate.

Lemma 4.1 (One-term exponential gap). *Let*

$$f(N) = \frac{\exp(2\sqrt{AN})}{4\sqrt{3}N} \left(1 - \frac{1}{2\sqrt{AN}}\right). \quad (35)$$

For positive integers n tending to infinity,

$$p(n) = f(n - 1/24) + O\left(\exp(\sqrt{A(n - 1/24)})\right). \quad (36)$$

The same estimate holds at each of finitely many fixed integer shifts of n .

Proof. Put $w = 2\sqrt{AN}$. The cited convergent series is

$$p(n) = \frac{1}{2\sqrt{3}N} \sum_{a \geq 1} \frac{A_a(n)}{\sqrt{a}} U(w/a), \quad U(x) = \cosh x - \frac{\sinh x}{x}. \quad (37)$$

Here $A_1(n) = 1$, and $A_a(n)$ is a sum of at most a unit-modulus terms, so $|A_a(n)| \leq a$. For $x \geq 0$,

$$U(x) = \frac{1}{x} \int_0^x u \sinh u \, du, \quad 0 \leq U(x) \leq \frac{x^2 e^x}{3},$$

where at $x = 0$ the formulas are understood by continuity. The bound follows from $\sinh u \leq ue^u \leq ue^x$. Hence the terms with $a \geq 2$ have total absolute value at most a constant times

$$N^{-1}w^2 e^{w/2} \sum_{a \geq 2} a^{-3/2} = O(e^{w/2}),$$

since $w^2/N = 4A$. The $a = 1$ term equals

$$\frac{1}{4\sqrt{3}N} \left(e^w(1 - w^{-1}) + e^{-w}(1 + w^{-1}) \right).$$

Its growing term is exactly (35), and its other term is exponentially small. This proves (36). Finitely many fixed shifts change w by $O(N^{-1/2})$, preserving the gap. $\square$

Lemma 4.2 (Exact derivative polynomials). *For every fixed $\ell \geq 0$,*

$$f^{(\ell)}(N) = B(N + 1/24)t^\ell Q_\ell(t), \quad t = \sqrt{A/N}, \quad (38)$$

where derivatives are with respect to the real variable N , and Q_ℓ is (10).

Proof. The case $\ell = 0$ follows from $Q_0(t) = 1 - t/(2A)$. Direct differentiation gives

$$\frac{B'(N + 1/24)}{B(N + 1/24)} = t - \frac{t^2}{A}, \quad \frac{dt}{dN} = -\frac{t^3}{2A}.$$

Thus (38) is equivalent to the recurrence

$$Q_{\ell+1}(t) = \left(1 - \frac{(\ell+2)t}{2A}\right) Q_\ell(t) - \frac{t^2}{2A} Q'_\ell(t). \quad (39)$$

To verify the terminating formula directly, write $Q_\ell(t) = \sum_h q_{\ell,h} t^h$ with coefficients outside $0 \leq h \leq \ell + 1$ set to zero. The recurrence is

$$q_{\ell+1,h} = q_{\ell,h} - \frac{\ell + h + 1}{2A} q_{\ell,h-1}.$$

Substituting the factorial expression in (10) satisfies this identity, including both endpoints. Induction proves the result. $\square$

Proposition 4.3 (Polynomial transfer). *Let $H(q) = \sum_d h_d q^d$ be a fixed polynomial. For every fixed nonnegative integer L ,*

$$[q^n]P(q)H(q) = B(n) \sum_{\ell=0}^L [z^\ell]H(e^{-z}) t^\ell Q_\ell(t) + O_{H,L}(B(n)t^{L+1}). \quad (40)$$

Proof. Coefficient extraction is the exact finite sum $\sum_d h_d p(n-d)$. For sufficiently large n , every $n-d$ in that sum is positive. Lemma 4.1 replaces $p(n-d)$ by $f(N-d)$ with an error exponentially smaller than $B(n)$ times any fixed power of t . Taylor's theorem at the finitely many bounded shifts d gives

$$f(N-d) = \sum_{\ell=0}^L \frac{(-d)^\ell}{\ell!} f^{(\ell)}(N) + O_{d,L}(B(n)t^{L+1}).$$

Indeed Lemma 4.2 bounds $f^{(L+1)}(N-u)$ by $O(B(n)t^{L+1})$ uniformly for u in the fixed interval containing these shifts. Summing and using

$$[z^\ell]H(e^{-z}) = \sum_d h_d \frac{(-d)^\ell}{\ell!}$$

proves (40). Only the explicit smooth function f is differentiated; no asymptotic error for integer partition numbers is differentiated. $\square$

Proof of Theorem 1.1. Choose J by (34), and let H be the finite sieve polynomial consisting of the first J terms in (19) or (20). Section 3.1 replaces $F_{k,b}^X$ by $P(q)H(q)$ with coefficient error $O(B(n)t^{s+R+1})$. Apply Proposition 4.3 with $L = s + R$.

The coefficients below degree s in $H(e^{-z})$ vanish exactly. Also the omitted $j > J$ terms have valuation at least

$$(k-1)(J+1) + \epsilon_X = s + (k-1)J > s + R,$$

so the coefficients through degree $s + R$ agree with those of $G_{k,b}^X$. Discarding total t -degrees beyond $s + R$ in (40) gives precisely (12) and (11). The $j = 1$ product has leading coefficient $\prod_{r=1}^k r = k!$ for equality, and $\prod_{r=2}^k r = k!$ for the tail. All $j \geq 2$ terms have strictly larger valuation. Therefore $g_s = k!$ and $c_0 = 1$.

Finally (36) gives $p(n) = B(n)(1 - t/(2A))$ plus an exponentially smaller term; $t \sim \pi/\sqrt{6n}$. This proves (13)–(14). $\square$

5 Coefficients and fixed offset companions

Formula (11) can be written without formal-series notation as

$$c_m = \frac{1}{k!} \sum_{\substack{s \leq \ell \leq s+m \\ 0 \leq h = s+m-\ell \leq \ell+1}} g_\ell \frac{(-1)^h (\ell + h + 1)!}{(\ell + 1 - h)! h! (4A)^h}. \quad (41)$$

The constrained sum avoids any negative-factorial convention. The finite upper bound for a purely algebraic coefficient calculation is

$$j \leq \left\lfloor \frac{s + R - \epsilon_X}{k - 1} \right\rfloor \quad (42)$$

when computing $g_0, \dots, g_{s+R}$. This algebraic bound differs from the larger analytical depth (34); the latter proves the error bound and cannot be replaced by a radial valuation argument.

5.1 The first correction and the exceptional second term

For the $j = 1$ term, $D_1 = k + b$ and

$$\prod_{r \in I} (1 - e^{-rz}) = \left(\prod_{r \in I} r \right) z^{|I|} \left(1 - \frac{z}{2} \sum_{r \in I} r + O(z^2) \right).$$

For equality $I = \{1, \dots, k\}$, giving a relative radial coefficient $-k(k+5)/4 - b$. For the tail $I = \{2, \dots, k\}$, it is $-[k(k+5) - 2]/4 - b$.

The $j = 2$ summand begins at relative degree $k - 1$. It enters the first correction only for $k = 2$. Its signed leading coefficients are $-(2 \cdot 3 \cdot 4) = -24$ for equality and $-(3 \cdot 4) = -12$ for the tail. Dividing by $k! = 2$ gives the extra -12 and -6 . Finally the coefficient of t in Q_s is $-(s+1)(s+2)/(4A)$. This proves (15)–(16). In particular, retaining only one selected large part captures the leading equivalent but loses a first-order contribution at slope 2.

Normalizing instead by the exact partition number gives

$$\frac{E_{k,b}(n)}{k!t^k p(n)} = 1 - \left(\frac{k(k+5)}{4} + b + 12\mathbf{1}_{\{k=2\}} + \frac{k(k+3)}{4A} \right) t + O(t^2), \quad (43)$$

$$\frac{T_{k,b}(n)}{k!t^{k-1} p(n)} = 1 - \left(\frac{k(k+5) - 2}{4} + b + 6\mathbf{1}_{\{k=2\}} + \frac{(k-1)(k+2)}{4A} \right) t + O(t^2). \quad (44)$$

At all fixed orders the conversion is multiplication by the formal reciprocal of $1 - t/(2A)$. The shift $N = n - 1/24$ is kept throughout, so these formulas should not be mixed termwise with coefficients in powers of $n^{-1/2}$ without re-expanding that shift.

The leading terms are independent of the fixed offset b , whereas the first relative coefficient changes by exactly $-b$. Identity (24) holds exactly. At higher orders the factors e^{-bjz} in the sieve determine the offset dependence algorithmically. This is a fixed-offset statement; summing these expansions over a growing range of offsets would require a new uniform remainder analysis.

5.2 The two OEIS specializations

For A237753, corresponding to $(k, b, X) = (2, 0, E)$, the radial coefficients begin

$$G_{2,0}^E(z) = 2z^2 - 31z^3 + \frac{2090}{3}z^4 - \frac{79007}{4}z^5 + \frac{30640414}{45}z^6 + \dots \quad (45)$$

Its first four normalized coefficients are

$$\begin{aligned} c_0 &= 1, \\ c_1 &= -\frac{31}{2} - \frac{3}{A}, \\ c_2 &= \frac{1045}{3} + \frac{155}{2A} + \frac{15}{4A^2}, \\ c_3 &= -\frac{79007}{8} - \frac{5225}{2A} - \frac{1395}{8A^2} - \frac{15}{8A^3}. \end{aligned} \quad (46)$$

They multiply $2B(n)t^2$ in (12).

For A237751, corresponding to $(k, b, X) = (2, 1, T)$,

$$G_{2,1}^T(z) = 2z - 20z^2 + \frac{931}{3}z^3 - \frac{20270}{3}z^4 + \frac{11331751}{60}z^5 + \dots, \quad (47)$$

and the coefficients multiplying $2B(n)t$ are

$$\begin{aligned} c_0 &= 1, \\ c_1 &= -10 - \frac{3}{2A}, \\ c_2 &= \frac{931}{6} + \frac{30}{A} + \frac{3}{4A^2}, \\ c_3 &= -\frac{10135}{3} - \frac{4655}{6A} - \frac{75}{2A^2}. \end{aligned} \quad (48)$$

The absence of an A^{-3} term in the last line is consistent with the terminating degree of Q_1 . These expansions are not obtained by merely substituting $k = 2$ into a formula that discards the $j = 2$ sieve term.

6 All fixed order inverse asymptotics

Fix one of the sequences in Theorem 1.1. For real $n > 1/24$ put

$$w = 2\sqrt{A(n - 1/24)}, \quad d = s + 2, \quad D = \frac{k! 2^s A^{s+1}}{\sqrt{3}}. \quad (49)$$

Then $t = 2A/w$ and the forward result is

$$X_{k,b}(n) = De^w w^{-d} \left(1 + \sum_{j=1}^R c_j \left(\frac{2A}{w} \right)^j + O(w^{-R-1}) \right) \quad (50)$$

on integer n . The expression outside the error is an explicit smooth function; the integer count is not being extended or differentiated.

6.1 Lambert core and triangular reversion

For sufficiently large real y , let

$$w_0 = -d W_{-1} \left(-\frac{1}{d} \left(\frac{D}{y} \right)^{1/d} \right). \quad (51)$$

This is the large positive solution of

$$De^{w_0} w_0^{-d} = y. \quad (52)$$

The branch W_{-1} , not W_0 , is required: if $y > D(e/d)^d$, then $w_0 > d$ and $w_0 \rightarrow \infty$ as $y \rightarrow \infty$.

Let $H(v) = 1 + \sum_{j \geq 1} c_j v^j$ be a formal series. Set $z = w_0^{-1}$, and seek

$$\Delta(z) = \sum_{j \geq 1} u_j z^j \quad (53)$$

so that substituting $w = w_0 + \Delta(z)$ in the logarithm of the forward expression yields zero residual. Using (52), the formal equation is

$$\Delta - d \log(1 + z\Delta) + \log H \left(\frac{2Az}{1 + z\Delta} \right) = 0. \quad (54)$$

All logarithms here are formal logarithms of series with constant term one.

Proposition 6.1 (Triangular inverse coefficients). *Equation (54) has a unique formal solution of the form (53). For every $m \geq 1$, its coefficient of z^m equals u_m plus a polynomial in $u_1, \dots, u_{m-1}$ and $c_1, \dots, c_m$, with coefficients rational in A and d . Thus u_m is determined by a finite triangular calculation. The first coefficients are*

$$u_1 = -2Ac_1, \quad (55)$$

$$u_2 = -2Adc_1 - 4A^2c_2 + 2A^2c_1^2, \quad (56)$$

$$u_3 = -\frac{8A^3}{3}c_1^3 + 8A^3c_1c_2 - 8A^3c_3 + 2A^2dc_1^2 - 4A^2c_1^2 \\ - 4A^2dc_2 - 2Ad^2c_1. \quad (57)$$

Proof. The direct term Δ contributes u_m to degree m . In $z\Delta$, the same unknown first appears at degree $m+1$, and in the substitution into H it can only appear at still higher degree. Therefore all other degree- m terms depend only on earlier u 's. The constant term vanishes; solving degree by degree proves existence and uniqueness. Expanding the two formal logarithms through degrees one, two, and three gives (55)–(57). $\square$

For each fixed $M \geq 1$ define the real center

$$\Delta_M = \sum_{j=1}^M \frac{u_j}{w_0^j}, \quad x_M(y) = \frac{1}{24} + \frac{(w_0 + \Delta_M)^2}{4A}. \quad (58)$$

The square in this definition is retained in full, even though some terms lie beyond its promised order. Define also the Lambert-only center $x_0 = 1/24 + w_0^2/(4A)$ for numerical comparison.

Lemma 6.2 (Error of the smooth inverse center). *Let*

$$\mathcal{F}_M(x) = De^{w(x)}w(x)^{-d} \left(1 + \sum_{j=1}^M c_j(2A/w(x))^j \right), \quad w(x) = 2\sqrt{A(x - 1/24)}. \quad (59)$$

On its large positive branch this function is positive and strictly increasing. If $\hat{x}_M$ is its large real solution of $\mathcal{F}_M(x) = y$, then

$$\hat{x}_M = x_M(y) + O(w_0^{-M}). \quad (60)$$

Proof. The polynomial in parentheses tends to one and is positive for large x . Its logarithmic derivative with respect to w is $O(w^{-2})$, so

$$\frac{d}{dx} \log \mathcal{F}_M(x) = \frac{2A}{w} \left(1 - \frac{d}{w} + O(w^{-2}) \right) = t + O(t^2) > 0 \quad (61)$$

for large x . This also gives a unique large root when y is large. Substitution of $w_0 + \Delta_M$ in the logarithmic equation leaves residual $O(w_0^{-M-1})$, by Proposition 6.1. The derivative of that equation with respect to w is $1 + O(w_0^{-1})$ in a bounded neighborhood. The mean value theorem therefore gives an error $O(w_0^{-M-1})$ in w . Since $dx/dw = w/(2A) = O(w_0)$, the error in x is $O(w_0^{-M})$. $\square$

In particular the first two index corrections, viewed as an expansion of the smooth inverse, are

$$\hat{x}_M = \frac{1}{24} + \frac{w_0^2}{4A} - c_1 + \frac{-dc_1 - 2Ac_2 + Ac_1^2}{w_0} + O(w_0^{-2}) \quad (M \geq 2). \quad (62)$$

For $M = 1$ the constant correction $-c_1$ is valid with error $O(w_0^{-1})$. Thus this constant is $31/2 + 3/A$ for A237753 and $10 + 3/(2A)$ for A237751. These constants improve a real center, not a universal rounding prescription.

6.2 Integer thresholds and the two ceilings

Use the exact threshold convention

$$\nu_X(y) = \min\{n \geq 0 : X_{k,b}(n) \geq y\}. \quad (63)$$

The inequality is weak; this matters at targets equal to a sequence value.

Lemma 6.3 (Eventual strict increase). *Each sequence in Theorem 1.1 is positive, unbounded, and strictly increasing for all sufficiently large integers n .*

Proof. Write the expansion through relative order one with remainder $O(t^2)$. The ratio of successive smooth leading factors $B(n)t(n)^s$ is $1 + t + O(t^2)$, by explicit differentiation or direct expansion of (5). The factors $1 + c_1 t + O(t^2)$ at n and $n + 1$ have ratio $1 + O(t^2)$ because $t(n + 1) - t(n) = O(t^3)$. Consequently

$$\frac{X_{k,b}(n+1)}{X_{k,b}(n)} = 1 + t + O(t^2) > 1 \quad (64)$$

for all sufficiently large n . The leading equivalent proves eventual positivity and unboundedness. This argument compares two error bounds; it never differentiates an unknown integer-index error. $\square$

Theorem 6.4 (Shrinking threshold uncertainty). *For every fixed $M \geq 1$ there are constants $C_M > 0$ and y_M such that, for every real $y \geq y_M$,*

$$\lceil x_M(y) - C_M w_0^{-M} \rceil \leq \nu_X(y) \leq \lceil x_M(y) + C_M w_0^{-M} \rceil. \quad (65)$$

The constants may depend on the fixed k, b, X, M . They are not computed effectively here.

Proof. Let $\hat{x}_M$ be the root from Lemma 6.2, and put $h = C w_0^{-M}$ with $C > 0$ to be chosen. Set

$$n_- = \lceil \hat{x}_M - h \rceil - 1, \quad n_+ = \lceil \hat{x}_M + h \rceil.$$

The first is strictly below $\hat{x}_M - h$, and the second is at least $\hat{x}_M + h$. Both are within a bounded distance of $\hat{x}_M$, since $h \rightarrow 0$. Their corresponding w values are $w_0 + O(w_0^{-1})$.

By (61), the logarithmic change of $\mathcal{F}_M$ between its root and either test integer has the required sign and magnitude at least a fixed positive constant times $h/w_0 = C w_0^{-M-1}$. The forward theorem at order M gives

$$X_{k,b}(n) = \mathcal{F}_M(n)(1 + O(w_0^{-M-1}))$$

uniformly at these two nearby integers; the factor in (59) is bounded away from zero there. Choose C large enough to dominate the two relative errors. Then

$$X_{k,b}(n_-) < y \leq X_{k,b}(n_+)$$

for large y . Eventual monotonicity excludes every integer at most n_- in the eventual increasing segment and includes n_+ . The finite initial segment is below y once y exceeds its maximum. Thus

$$\lceil \hat{x}_M - h \rceil \leq \nu_X(y) \leq \lceil \hat{x}_M + h \rceil.$$

Finally (60) lets us enlarge the constant and replace $\hat{x}_M$ by x_M . This proves (65) with the exact weak threshold convention (63). $\square$

For sufficiently large y the interval in (65) has width less than one, so the two ceilings are equal or adjacent. When they agree they identify the threshold exactly. When x_M lies too close to an integer, an unconditional assertion $\nu_X(y) = \lceil x_M(y) \rceil$ is not justified. The theorem states an existential asymptotic bracket, not a certified finite- y enclosure. In particular numerical errors in a few examples cannot be used to select a valid universal C_M .

Remark 6.5 ([write] The inverse and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* By Lemma 6.3, for y beyond the finite initial segment $\nu_X(y)$ of (63) is the integer staircase $N_*(y)$ of `p0:def:three-inverses`, so `p0:thm:staircase(1)` applies. In the counts computed by the write, $E_{2,0}$ increases strictly from $n = 15$ and $T_{2,1}$ from $n = 5$ through $n = 3000$; that is a finite observation, not a proof of the onset.

- (b) *Instance.* The Lambert root w_0 of (51) is `p0:thm:lambert-core(3)`, case $b_{\text{vol}} < 0$, with $X_{\text{vol}} = w_0$, $a_{\text{vol}} = 1$, $b_{\text{vol}} = -d$, $L_{\text{vol}} = \log(y/D)$: the equation (52) is $w_0 - d \log w_0 = L_{\text{vol}}$, and the large solution $(|b_{\text{vol}}|/a_{\text{vol}})(-W_{-1}(-(a_{\text{vol}}/|b_{\text{vol}}|)e^{-L_{\text{vol}}/|b_{\text{vol}}|}))$ is (51); the condition $y > D(e/d)^d$ is the fold condition $L_{\text{vol}} > L_c$.
- (c) *Formal instance.* The reversion equation (54) is the formal-exactness identity of `p0:thm:lambert-centered(4)` with $a_{\text{vol}} = 1$, $b_{\text{vol}} = -d$, $Q_{\text{vol}}(\tau) = \log H(2A\tau)$, $\tau = 1/w_0$ and $U = \Delta$: with $x = w$, $a(x - \tau^{-1}) + b \log(\tau x) + Q(1/x) = \Delta - d \log(1 + z\Delta) + \log H(2Az/(1 + z\Delta))$. So Proposition 6.1 is an instance, u_m is the volume's c_m , and the volume's Bell recurrence `p0:eq:c-bell` reproduces (55)–(57) (checked by solving the reversion through degree three).
- (d) *Instance.* Put $Q_M(\tau) = \log(1 + \sum_{j \leq M} c_j (2A\tau)^j)$, whose first M coefficients are those of the full series. In the variable w , the function $\mathcal{F}_M$ of (59) satisfies

$$\log \mathcal{F}_M = \log D + w - d \log w + Q_M(1/w)$$

exactly, so it is of exponential–power type (`p0:def:model`) with data $(D, 1, 1, -d, 1, Q_M)$. Hence the root estimate behind (60) is `p0:thm:lambert-centered(5)` in w , followed by $x = \frac{1}{24} + w^2/(4A)$; its residual-over-slope step is `p0:thm:backward-error`.

- (e) *Analogue.* Theorem 6.4 is, as stated and proved, an analogue of `p0:thm:staircase(2)`: a two-ceiling bracket proved directly at the integers, not deduced from an admissible interpolation.

No novelty is claimed for any of these inversions.

7 Exact checks and numerical diagnostics

The companion package separates exact arithmetic checks from floating-point illustrations. The former verify finite identities and coefficient implementations; the analytic remainder and threshold theorems rest on the proofs above.

7.1 Independent finite paths

The public exact core uses Python's standard library, including arbitrary-precision integers and rational fractions. Partition numbers can be generated by Euler's pentagonal recurrence. For exact statistic values, each sieve summand is formed as a finite product and convolved with that partition sequence. It retains all j for which $D_j \leq n_{\text{max}}$, so no asymptotic approximation enters these counts.

A separate bounded-part, exact-length dynamic program supplies an independent counting path. Before adding part size M , its entries count partitions with all parts less than M . The ordinary unbounded coin-change update adds partitions whose maximum is exactly M , keeping the length coordinate. Each newly added contribution is assigned to every selected relation $M = kL + b$ or $M \geq kL + b$. Summing over all maxima and lengths recovers an independently computed unrestricted partition sequence. This checks the sieve without reusing its inclusion–exclusion identity.

For coefficient checks, one path multiplies truncated exponential series with rational coefficients. A different path first constructs the finite polynomial $H(q) = \sum_d h_d q^d$, then computes moments $\sum_d h_d (-d)^\ell / \ell!$. The transfer is also checked by comparing the terminating factorial formula with the differential recurrence (39). The inverse generator verifies its formal residual to the requested order. All guards are explicit exceptions and remain active with Python's `-O` flag.

The saved exact checks compare all 98 parameter and kind combinations with $2 \leq k \leq 8$, $0 \leq b \leq 6$, and $0 \leq n \leq 160$, totaling 15,778 integer comparisons. They check 72 moment, transfer, and inverse cases through relative order six, the Q_ℓ recurrence through $\ell = 20$, and 690 sufficient-depth combinations. There are 23 invalid-input and poisoned-receipt rejection controls, with an additional 26 package-integrity rejection controls. These figures describe finite checks, not the scope of the theorem. The independent algorithms were adapted from separate research-stage calculations; no claim of clean-room authorship or independent proof is made.

7.2 How to read the diagnostics

For a forward truncation order R , define

$$\mathcal{A}_R(n) = k!B(n)t^s \sum_{m=0}^R c_m t^m, \quad \text{relerr}_R(n) = \frac{\mathcal{A}_R(n)}{X_{k,b}(n)} - 1.$$

For inverse order M , the diagnostic is $x_M(X_{k,b}(n)) - n$, using the real center (58). Order zero uses the Lambert-only center. These are signed errors, not rigorous enclosures. The integer target is generated exactly; the subsequent transcendental evaluations use optional high-precision arithmetic. The diagnostic code records the precision and library version. At each of the 16 displayed targets $y = X_{k,b}(n)$, a scan of every exact count from zero through n , together with adjacent comparisons, verifies $\nu_X(y) = n$. This is a finite verification of these particular thresholds, independent of the asymptotic inverse errors.

X	(k, b)	n	0	1	2	3	6
E	(2, 0)	1000	6.262×10^{-1}	-5.164×10^{-1}	5.452×10^{-1}	-7.057×10^{-1}	3.930×10^0
E	(2, 0)	10000	2.105×10^{-1}	-5.842×10^{-2}	2.060×10^{-2}	-8.845×10^{-3}	1.741×10^{-3}
T	(2, 0)	1000	3.487×10^{-1}	-1.935×10^{-1}	1.534×10^{-1}	-1.586×10^{-1}	5.469×10^{-1}
T	(2, 0)	10000	1.196×10^{-1}	-2.271×10^{-2}	6.085×10^{-3}	-2.105×10^{-3}	2.612×10^{-4}
E	(2, 1)	1000	6.749×10^{-1}	-5.699×10^{-1}	6.091×10^{-1}	-7.942×10^{-1}	4.471×10^0
E	(2, 1)	10000	2.242×10^{-1}	-6.352×10^{-2}	2.264×10^{-2}	-9.787×10^{-3}	1.946×10^{-3}
T	(2, 1)	1000	3.956×10^{-1}	-2.220×10^{-1}	1.767×10^{-1}	-1.831×10^{-1}	6.333×10^{-1}
T	(2, 1)	10000	1.331×10^{-1}	-2.551×10^{-2}	6.859×10^{-3}	-2.377×10^{-3}	2.958×10^{-4}
E	(3, 0)	1000	5.491×10^{-1}	-1.885×10^{-2}	-1.970×10^{-1}	1.448×10^{-1}	1.235×10^{-1}
E	(3, 0)	10000	1.382×10^{-1}	6.249×10^{-3}	-6.839×10^{-3}	1.102×10^{-3}	2.349×10^{-5}
T	(3, 0)	1000	3.925×10^{-1}	-2.110×10^{-2}	-8.118×10^{-2}	5.522×10^{-2}	3.361×10^{-2}
T	(3, 0)	10000	1.057×10^{-1}	1.812×10^{-3}	-2.958×10^{-3}	4.666×10^{-4}	7.834×10^{-6}
E	(2, 3)	1000	7.783×10^{-1}	-6.876×10^{-1}	7.544×10^{-1}	-9.998×10^{-1}	5.767×10^0
E	(2, 3)	10000	2.520×10^{-1}	-7.437×10^{-2}	2.715×10^{-2}	-1.191×10^{-2}	2.421×10^{-3}
T	(2, 3)	1000	4.952×10^{-1}	-2.878×10^{-1}	2.320×10^{-1}	-2.420×10^{-1}	8.454×10^{-1}
T	(2, 3)	10000	1.605×10^{-1}	-3.172×10^{-2}	8.625×10^{-3}	-3.009×10^{-3}	3.777×10^{-4}

Table 1: Forward relative error $\mathcal{A}_R(n)/X(n) - 1$ at fixed truncation orders. These 80-digit mpmath evaluations are non-certified diagnostics, not error bounds.

Large alternating coefficients make these fixed-order expansions numerically demanding. At a moderate index, increasing the truncation order may make an approximation worse. Such behavior does not conflict with Theorem 1.1, which fixes the order before n tends to infinity, and does not establish an optimal truncation or resummation rule. The package includes fixed-offset and slope-3 companions to check that the implementation is not specialized only to the two named sequences.

X	(k, b)	n	0	1	2	3	6
E	(2, 0)	1000	-1.257×10^1	4.826×10^0	-4.446×10^0	5.653×10^0	-3.644×10^1
E	(2, 0)	10000	-1.513×10^1	2.203×10^0	-6.962×10^{-1}	2.966×10^{-1}	-6.788×10^{-2}
T	(2, 0)	1000	-7.645×10^0	2.292×10^0	-1.722×10^0	1.810×10^0	-7.266×10^0
T	(2, 0)	10000	-8.912×10^0	1.002×10^0	-2.582×10^{-1}	9.136×10^{-2}	-1.312×10^{-2}
E	(2, 1)	1000	-1.333×10^1	5.075×10^0	-4.698×10^0	6.008×10^0	-3.932×10^1
E	(2, 1)	10000	-1.601×10^1	2.318×10^0	-7.361×10^{-1}	3.153×10^{-1}	-7.324×10^{-2}
T	(2, 1)	1000	-8.517×10^0	2.425×10^0	-1.838×10^0	1.945×10^0	-7.931×10^0
T	(2, 1)	10000	-9.853×10^0	1.062×10^0	-2.758×10^{-1}	9.819×10^{-2}	-1.431×10^{-2}
E	(3, 0)	1000	-1.147×10^1	-2.409×10^0	2.700×10^0	-1.092×10^0	-9.135×10^{-1}
E	(3, 0)	10000	-1.029×10^1	-1.250×10^0	3.482×10^{-1}	-2.479×10^{-2}	-2.171×10^{-4}
T	(3, 0)	1000	-8.570×10^0	-1.233×10^0	1.301×10^0	-5.087×10^{-1}	-3.261×10^{-1}
T	(3, 0)	10000	-7.955×10^0	-6.295×10^{-1}	1.657×10^{-1}	-1.317×10^{-2}	-1.767×10^{-4}
E	(2, 3)	1000	-1.488×10^1	5.553×10^0	-5.223×10^0	6.762×10^0	-4.564×10^1
E	(2, 3)	10000	-1.779×10^1	2.543×10^0	-8.195×10^{-1}	3.550×10^{-1}	-8.499×10^{-2}
T	(2, 3)	1000	-1.027×10^1	2.680×10^0	-2.082×10^0	2.235×10^0	-9.410×10^0
T	(2, 3)	10000	-1.174×10^1	1.179×10^0	-3.129×10^{-1}	1.129×10^{-1}	-1.693×10^{-2}

Table 2: Inverse real-index error $x_M(X(n)) - n$; order zero is the Lambert core. These 80-digit mpmath evaluations are non-certified diagnostics, not error bounds.

8 Relation to prior partition asymptotics

This section records a bounded comparison of the cited source statements, their parameters, and their errors. It is not an exhaustive historical survey or a priority certificate. The principal analytic inputs in the proof are classical eta inversion and the exact Rademacher series. The model-specific argument is the finite-product sieve with its global finite-depth remainder and its explicit fixed-slope coefficients and inverse.

8.1 Joint extreme counts and rectangles

Szekeres’s *Asymptotic distribution of partitions by number and size of parts* [6] uses $p(n; r, k)$ for the exact joint count with length r and largest part k . Its Theorem 1 sets $c = \pi/\sqrt{6}$, $\lambda = ck/\sqrt{n}$, and $\mu = cr/\sqrt{n}$, and assumes

$$\frac{1}{4} \log n < \lambda, \mu < \frac{n^{1/4}}{\log n}.$$

The resulting leading equivalent contains correlation terms in the exponent. A corollary on printed page 530 gives an independent-extreme simplification under narrower assumptions. The proof on printed pages 532–537 keeps an $o(1)$ exponential remainder. Thus both the broad correlated theorem and the narrower independent-extreme law are relevant prior work. The original exact-count definition matters; a later paper’s cumulative notation should not be substituted for it. These leading joint asymptotics do not themselves provide the arbitrary fixed-order diagonal sum and offset calculus proved here.

Jiang and Wang [7] give a generalized Hardy–Ramanujan formula for cumulative rectangular restrictions. Their Theorem 1 and Remarks 1.2–1.3 allow both cutoffs to be at least $4\sqrt{n}$, with unbounded normalized cutoffs, and have relative error $O(n^{-1/4+\epsilon})$. This range overlaps the $\sqrt{n} \log n$ cutoff sizes relevant to fixed-slope extreme events. It would be incorrect to dismiss their theorem as confined to bounded multiples of $\sqrt{n}$. Its stated error, however, does not justify taking the rare diagonal differences and sums to every algebraic order.

Melczer, Panova, and Pemantle [8] treat partitions inside a rectangle. Their Theorems 1–2 require the normalized parameters $(\ell/m, n/m^2)$ to remain in a compact subset of $x \geq 2y > 0$. For m of order $\sqrt{n} \log n$, the second coordinate tends to zero, outside that compact regime. Their refined consecutive- n differences also hold the rectangle fixed. These are important rectangle results, but their particular compact-uniform theorem does not directly cover the full dominant fixed-slope sum.

Richmond [9] treats central extreme coordinates and extended one-sided cumulative ranges. The central window $|x_i| \leq (\log n)/5$ has a displayed error $O(n^{-1/10} \log n)$. Equation (10) also treats exact joint counts, so the work should not be described as cumulative-only. It includes successive-rank questions. Its stated leading-error framework does not supply the all-fixed-order slope-coupled expansion or inverse threshold result here.

8.2 Sieve and transfer antecedents

Hwang [10] develops an analytic sieve for partitions with many summands. Theorem 1 compares $p(n, m)$ with $p(n - m)$ in a large-length regime, with a stated relative power error; Proposition 1 gives a general finite sieve with a controlled remainder. Later theorems extend to restricted part sets and length laws. This is a substantive methodological antecedent. It does not state the summed $M = kL + b$ or $M \geq kL + b$ expansion and inverse treated here.

Ngo and Rhoades [11] establish arbitrary-order expansions for partition statistics in a quantum-modular framework, with general amplitude-transfer results and polynomial and logarithmic variants. Therefore a claim that partition-product amplitudes admit arbitrary-order expansions would be too broad as a novelty claim. Our elementary finite-shift transfer makes the particular coefficient normalization transparent, while the substantive analytic obligation is the global remainder for this fixed-slope amplitude.

Johansson [12] provides the exact Rademacher normalization used in Lemma 4.1, including a truncation estimate that tends to zero. We proved the one-term exponential gap from that source’s convergent formula, rather than merely quoting a leading equivalent. The classical original is Rademacher [13]; the directly inspected and used formula is Johansson’s. Eta inversion is taken in the normalization recorded by the DLMF [14].

8.3 Rank terminology and the excluded slope

At $k = 1$, $M - L$ is Dyson’s ordinary rank. Dyson’s exact alternating identities and subsequent rank literature are prior work [16]. The finite-depth gain in Proposition 3.2 is $k - 1$, which vanishes at that slope. No $k = 1$ asymptotic conclusion is claimed here.

The named k -rank discussed by Berkovich and Garvan [17, equations (6.1) and (6.2)] is defined using successive Durfee squares, selected columns, and rows. It is not the statistic $M - kL$. We use the descriptive phrase “fixed-slope extreme statistic” to avoid identifying these different notions. A finite search of rank terminology cannot exclude an unindexed treatment of the same diagonal event elsewhere.

9 Reproducibility and scope of the package

The compact archive contains the editable L^AT_EX manuscript, its PDF, standard-library exact code, selected generated rational coefficients and exact counts, optional numerical diagnostics, the

generated tables, source references, and a checksum manifest. It does not redistribute source papers. No third-party source PDF is required to rerun the exact finite checks, and no network access is required for regeneration.

The exact core and its tests remain usable without the optional numerical library. The full PDF-and-archive build additionally needs the documented T_EX toolchain and the optional library for the diagnostic tables. The entry-point README gives commands for a fresh output directory, normal and optimized execution, byte-level verification of regenerated artifacts, and package guard tests. Same-toolchain byte stability is a reproducibility property of these files; it is not a claim that different T_EX, font, Python, or numerical-library versions necessarily emit identical bytes.

Finite tests cover only the ranges explicitly recorded in their receipts. Agreement of independently generated finite counts supports the implementation and the exact identities, but cannot prove the asymptotic remainder. Numerical diagnostics support neither effective constants in (12) nor a certified finite-target enclosure in (65). The latter remain theorem statements with existential constants.

10 Further directions

Several extensions would require additional arguments rather than a change of notation.

1. An effective version of the global bounds, the Rademacher tail estimate, and the finite-shift remainder could in principle produce explicit constants and an onset for a chosen k, b, R . That work would turn the inverse bracket into a computable enclosure. It is not supplied by the present diagnostic tables.
2. Allowing b or k to vary with n changes the uniformity of the numerator bounds and the finite polynomial retained in the transfer. In particular a growing shift is no longer a bounded Taylor shift. The fixed-parameter theorem cannot simply be reused.
3. The coefficients grow rapidly in the slope-2 examples. Understanding optimal truncation, large-order coefficient behavior, and exponentially small corrections would be a distinct problem. No convergent resummation follows from the all-fixed-order expansion.
4. The boundary $k = 1$ belongs to the ordinary rank problem, with no positive sieve-valuation gain. A uniform transition towards that regime would need a different analytic organization and a careful comparison with existing rank asymptotics.
5. Similar finite-product identities may arise for other coupled diagram boundaries. Each model would need its own entire-circle majorant; a formal radial series is not a substitute for that bound.

[write] **7 October 2026.** Under Vladimir's standing rule of 4 October 2026 (unproved claims are moved to questions, wrong ones are refuted on the record), the write found no claim of this Part that is unproved as stated and none that is wrong. The one wrong claim met in its sources is the OEIS alternative form discussed in Section 1; it is refuted, with proof, in Remark 1.3. On Item 1, note only that the write's exact counts increase strictly from $n = 15$ ($E_{2,0}$) and $n = 5$ ($T_{2,1}$) through $n = 3000$, which proves no onset.

A Exact algorithms in finite arithmetic

This appendix records the finite calculations independently of any particular programming language.

Exact sequence values. To compute $X_{k,b}(0), \dots, X_{k,b}(n_{\max})$, first compute $p(0), \dots, p(n_{\max})$. For each j with $D_j \leq n_{\max}$, start with the truncated coefficient list of $P(q)$. Multiply it successively by $1 - q^r$ for $r = j, \dots, kj$ in the equality case, or $r = j + 1, \dots, kj$ for the tail. For a single factor, update coefficients in descending index order by $a_m \leftarrow a_m - a_{m-r}$. Shift by D_j , multiply by $(-1)^{j-1}$, and add to the output. All arithmetic is integral. Descending order is essential: ascending order would implement a different operation.

Rational forward coefficients. For the requested relative order R , set $L = s + R$. For each j allowed by (42), multiply the truncated series

$$e^{-D_j z} = \sum_{a=0}^L \frac{(-D_j)^a}{a!} z^a + O(z^{L+1}), \quad 1 - e^{-rz} = - \sum_{a=1}^L \frac{(-r)^a}{a!} z^a + O(z^{L+1}).$$

Add the signed product to obtain $g_0, \dots, g_L$. Then use (41). Every operation is rational, and A^{-1} is an indeterminate until numerical evaluation is explicitly requested. This exact construction also verifies $g_0 = \dots = g_{s-1} = 0$ and $g_s = k!$.

Independent coefficient moments. Build the finite sieve polynomial $H(q)$ in integer arithmetic instead. If $H(q) = \sum_d h_d q^d$, then

$$g_\ell = \frac{1}{\ell!} \sum_d h_d (-d)^\ell \quad (0 \leq \ell \leq L).$$

Compute Q_ℓ by (39), rather than by its factorial formula. This path compares two different finite constructions of both the radial and transfer coefficients.

Triangular inverse. Suppose $u_1, \dots, u_{m-1}$ have been found. Set $u_m = 0$, substitute the truncated Δ into the left side of (54), and extract its degree- m coefficient r_m . Then set $u_m = -r_m$. Formal multiplication, reciprocal, and logarithm are all truncated at degree m . The inverse coefficients can be stored as rational Laurent polynomials in A , since the forward coefficients are rational polynomials in A^{-1} . The residual through the requested degree must vanish exactly. A negative control that changes one coefficient must fail that residual test.

Part II

Exact ratio two partitions at fixed asymptotic order

Bundle Report 217, archive Report217-source-v2.zip, delivered as `article.tex` (14 pages). Delivered title block “Exact ratio two partitions at fixed asymptotic order / Report 217 / 4 October 2026”. Printed as delivered, with `[write]` remarks and notes; Report 217’s Section k is Section $k+10$ here and its equation (k) is equation ($k+65$); statements move with their sections. Its labels carry `xpc:r2`; its three unlabelled sections were labelled; its citation keys `oeis` and `dlmf` are `oeisA118096` and `dlmfBesse1`. This Part is self-contained.

Abstract of this Part (as delivered). Let $a(n)$ count partitions of n whose largest part is exactly twice their smallest part, the sequence OEIS A118096. We give a self-contained fixed-order coefficient expansion, an exact finite rule for every correction, the first four corrections, and rounding-aware inverse bounds. A complete-circle estimate suppresses every noncentral angle, and a discrete two-variable saddle provides a uniform finite-order remainder. This is an independently derived refinement, not a claim of novelty or a solution of a known open problem. The leading equivalent was already recorded by Kotěšovec in 2025; general radial all-order methods precede this account. Historical comparison is incomplete: Pittel's directly adjacent 2007 bounded-ratio paper was identified, but its full theorem text was not accessible for this comparison. That limitation is not evidence of absence of overlapping results.

11 Scope and main results

A partition has positive integer parts. For $n \geq 0$, let $a(n)$ count those with largest part $2k$ and smallest part k for some $k \geq 1$; in particular $a(0) = 0$. Requiring one part of each endpoint gives, for $|q| < 1$,

$$F(q) = \sum_{n \geq 0} a(n)q^n = \sum_{k \geq 1} T_k(q), \quad T_k(q) = \frac{q^{3k}}{\prod_{j=k}^{2k} (1 - q^j)}. \quad (66)$$

The sum converges locally absolutely in the disk. The endpoint copies account for $3k$; all additional parts lie in $[k, 2k]$. Set

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad x_0 = \log \phi, \quad A = \frac{\pi^2}{30}, \quad B = 2\sqrt{A}, \quad C = \frac{1}{\sqrt{5 + \sqrt{5}}}. \quad (67)$$

Theorem 11.1 (Coefficient expansion). *For every fixed integer $M \geq 0$, as integer $n \rightarrow \infty$,*

$$a(n) = Cn^{-1/2}e^{B\sqrt{n}} \left(\sum_{r=0}^M d_r n^{-r/2} + O_M(n^{-(M+1)/2}) \right), \quad d_0 = 1. \quad (68)$$

The constants are determined by the exact finite rule in Section 15 and the transfer formula

$$d_r = \sum_{j=\lceil r/2 \rceil}^r c_j A^{j/2} \frac{(-1)^{r-j} r!}{(r-j)!(2j-r)!(4\sqrt{A})^{r-j}}. \quad (69)$$

In particular, $c_0 = 1$ and

$$\begin{aligned} c_1 &= \frac{239}{120} + \sqrt{5}, \\ c_2 &= \frac{232801}{28800} + \frac{431\sqrt{5}}{120}, \\ c_3 &= \frac{640181519}{10368000} + \frac{795361\sqrt{5}}{28800}, \\ c_4 &= \frac{3268837156801}{4976640000} + \frac{3045555791\sqrt{5}}{10368000}. \end{aligned} \quad (70)$$

Consequently

$$\begin{aligned} d_1 &= \sqrt{A} c_1, & d_2 &= A c_2 - \frac{1}{2} c_1, \\ d_3 &= A^{3/2} c_3 - \frac{3}{2} \sqrt{A} c_2, & d_4 &= A^2 c_4 - 3A c_3 + \frac{3}{4} c_2. \end{aligned} \quad (71)$$

Numerically these are 2.424917491063, 3.187616630712, 9.439637377883, and 32.379308690155, respectively.

The $M = 0$ equivalent is the one already recorded in A118096 [4]:

$$\frac{\exp(\pi\sqrt{2n/15})}{5^{1/4}\sqrt{2\phi n}}.$$

All O -constants in this article may depend on the fixed requested order. Neither uniformity as that order grows nor an explicit numerical threshold is asserted. The coefficients are computable exactly; the asymptotic remainder constants are not certified by the accompanying numerical experiments.

The proof separates two issues that should not be conflated. The real-axis expansion belongs to the existing Eulerian-series framework of McIntosh [18] and Zhou [19]. It cannot by itself be transferred to all coefficient orders by a leading-term Tauberian argument. Sections 13–16 give the angular estimate, the uniform two-variable remainder, and a finite Gaussian-contour identification needed here. Section 17 then gives a standard asymptotic inversion with two ceilings, and Section 18 proves eventual strict increase and log-concavity without differentiating an uncontrolled remainder.

Remark 11.2 ([write] The OEIS entry, quoted, 7 October 2026). OEIS A118096, read on 7 October 2026 in the internal format, is at revision #57 (18 March 2026), author Emeric Deutsch (12 April 2006), offset 1. Its name is “Number of partitions of n such that the largest part is twice the smallest part.”; its formula lines are “G.f.: $\text{Sum}_{\{k \geq 1\}} x^{(3 \cdot k)} / \text{Product}_{\{j = k..2 \cdot k\}} (1 - x^j)$.” (this Part’s (66)) and Kotěšovec’s “ $a(n) \sim \exp(\text{Pi} \cdot \text{sqrt}(2 \cdot n/15)) / (5^{(1/4)} \cdot \text{sqrt}(2 \cdot \phi \cdot n))$ ” (13 June 2025), which is the case $M = 0$ of Theorem 11.1 ($5^{1/4}\sqrt{2\phi} = \sqrt{5 + \sqrt{5}}$); a comment gives the conjugate reading (the number of parts is twice the multiplicity of the largest part). It also links a 2025 preprint of Stephen McKean, which this Part does not cite and the write did not read. Its b-file runs to $n = 10000$ (Kotěšovec; terms to 1000 from Alois P. Heinz). The bibliography entry [4] prints the name as a title in its own words. The write computed $a(n)$ from (66) for $n \leq 3000$: every b-file term in that range agrees, as do brute-force counts for $n \leq 40$. A118096 is cross-referenced from A117086, whose partitions with $\max / \min = 2$ are exactly these (Part III). Nothing was submitted to the OEIS.

12 The radial saddle

Define, on the positive real axis,

$$\begin{aligned} f(x) &= -\log(1 - e^{-x}), & h(x) &= \int_x^{2x} f(u) du = \text{Li}_2(e^{-x}) - \text{Li}_2(e^{-2x}), \\ g(x) &= \frac{e^{-3x}}{\sqrt{(1 - e^{-x})(1 - e^{-2x})}}. \end{aligned} \tag{72}$$

Their analytic continuations are taken only in a fixed small complex neighborhood of x_0 , with the branches determined by their positive real values. Put

$$a = -h''(x_0) = 3\phi - 4 > 0, \quad \lambda = g(x_0)\sqrt{\frac{2\pi}{a}} = \sqrt{\frac{2\pi}{\sqrt{5}\phi}}. \tag{73}$$

The symbol a without an argument denotes this positive saddle curvature; $a(n)$ always denotes the partition count.

Lemma 12.1. *The function h has a unique maximum on $(0, \infty)$, at x_0 . Its value is $A = \pi^2/30$, and*

$$D := 2A - ax_0^2 = \int_{x_0}^{2x_0} u^2 f''(u) du > 0. \tag{74}$$

Proof. With $y = e^{-x}$,

$$h'(x) = 2f(2x) - f(x) = -\log((1-y)(1+y)^2).$$

The equation $h'(x) = 0$ reduces to $y^2 + y = 1$. The sign is positive before x_0 and negative after it. Also $h(x) \rightarrow 0$ at both endpoints of $(0, \infty)$, and direct differentiation gives $h''(x_0) = 4 - 3\phi$.

For completeness, put $r = \phi^{-1}$ and $L = \log \phi$, so $r + r^2 = 1$. The dilogarithm identities needed here are

$$\begin{aligned} \text{Li}_2(r) + \text{Li}_2(r^2) &= \pi^2/6 - 2L^2, \\ \text{Li}_2(r) + \text{Li}_2(-r) &= \frac{1}{2} \text{Li}_2(r^2), \\ \text{Li}_2(r^2) &= -\text{Li}_2(-r) - \frac{1}{2}L^2. \end{aligned}$$

They are respectively reflection, duplication, and the identity $\text{Li}_2(x/(1+x)) = -\text{Li}_2(-x) - \frac{1}{2} \log^2(1+x)$ at $x = r$. Each follows by differentiation from $\text{Li}_2'(z) = -\log(1-z)/z$, with the constants obtained at zero or from $\text{Li}_2(1) = \sum_{m \geq 1} m^{-2} = \pi^2/6$. Solving gives $\text{Li}_2(r) = \pi^2/10 - L^2$ and $\text{Li}_2(r^2) = \pi^2/15 - L^2$, proving $h(x_0) = A$. Finally, twice integrating by parts gives

$$\int_x^{2x} u^2 f''(u) du = x^2 h''(x) - 2x h'(x) + 2h(x).$$

At the saddle this is D ; positivity follows from $f''(u) = e^{-u}/(1 - e^{-u})^2 > 0$. $\square$

Lemma 12.2 (Radial tails). *Fix $0 < \varepsilon < x_0 < R$, with $\varepsilon \leq 1$, and put $\eta = A - \max\{h(\varepsilon), h(R)\} > 0$. Uniformly for $0 < t \leq 1$ and $-\pi \leq \theta \leq \pi$,*

$$\sum_{kt \notin [\varepsilon, R]} |T_k(e^{-t+i\theta})| = O_{\varepsilon, R}(t^{-2} e^{(A-\eta)/t}). \quad (75)$$

Moreover $F(e^{-t}) = O(t^{-2} e^{A/t})$.

Proof. Since f is decreasing,

$$\sum_{j=k}^{2k} f(jt) \leq t^{-1} h(kt) + f(kt), \quad T_k(e^{-t}) \leq \frac{e^{h(kt)/t - 3kt}}{1 - e^{-kt}}.$$

For $k < \varepsilon/t$ use $1 - e^{-kt} \geq 1 - e^{-t} \geq t/2$ and at most ε/t terms. For $k > R/t$ use $1 - e^{-kt} \geq 1 - e^{-R}$ and sum the geometric factor e^{-3kt} , whose sum is $O(t^{-1})$. Monotonicity of h on the two tails supplies the exponent gap. On the compact interval each summand is at most a constant times $e^{A/t}$ and there are $O(t^{-1})$ terms. Finally $|1 - q^j| \geq 1 - |q|^j$ implies $|T_k(q)| \leq T_k(|q|)$. $\square$

13 A complete circle estimate

The following elementary bound controls rational and irrational angles at once.

Lemma 13.1. *For an integer $L \geq 2$, an integer k , and $|\theta| \leq \pi$,*

$$\sum_{j=k}^{k+L-1} (1 - \cos(j\theta)) \geq \frac{L}{81\pi^2} \min\{L^2\theta^2, 1\}. \quad (76)$$

Proof. Let $U = \sum_{j=0}^{L-1} e^{ij\theta}$. The left side is at least $L - |U|$, independently of k . For $|\theta| \leq 2\pi/L$ set $m = \lfloor L/2 \rfloor$. The identity

$$L^2 - |U|^2 = 2 \sum_{d=1}^{L-1} (L-d)(1 - \cos(d\theta))$$

and $1 - \cos u \geq 2u^2/\pi^2$ for $|u| \leq \pi$ imply

$$L - |U| \geq \frac{2\theta^2}{\pi^2 L} \sum_{d=1}^m (L-d)d^2 \geq \frac{L^3\theta^2}{81\pi^2}.$$

Here $L - d \geq L/2$, $m \geq L/3$, and $\sum_{d=1}^m d^2 \geq m^3/3$ suffice. If $|\theta| \geq 2\pi/L$, the geometric sum gives $|U| \leq 1/|\sin(\theta/2)| \leq \pi/|\theta| \leq L/2$. These two cases imply the stated, slightly weaker, bound. $\square$

For $\varepsilon \leq kt \leq R$, write $r_j = e^{-jt}$. Exactly,

$$\frac{|T_k(e^{-t+i\theta})|}{T_k(e^{-t})} = \prod_{j=k}^{2k} \left(1 + \frac{2r_j(1 - \cos(j\theta))}{(1 - r_j)^2} \right)^{-1/2}.$$

For $0 \leq u \leq 2$ and $r \geq e^{-2R}$, concavity of the logarithm on this interval gives

$$\frac{1}{2} \log \left(1 + \frac{2ru}{(1-r)^2} \right) \geq u \operatorname{arctanh}(r) \geq \gamma_R u, \quad \gamma_R = \operatorname{arctanh}(e^{-2R}) > 0.$$

Indeed the chord slope from $u = 0$ to $u = 2$ is $\frac{1}{4} \log(1 + 4r/(1-r)^2) = \operatorname{arctanh}(r)$. Apply Lemma 13.1 with $L = k + 1 \geq \varepsilon/t$. Since $\varepsilon \leq 1$,

$$|T_k(e^{-t+i\theta})| \leq T_k(e^{-t}) \exp \left[-c \min \left\{ \frac{\theta^2}{t^3}, \frac{1}{t} \right\} \right], \quad c = \frac{\gamma_R \varepsilon^3}{81\pi^2} > 0. \quad (77)$$

Summation and Lemma 12.2 prove

$$|F(e^{-t+i\theta})| \leq F(e^{-t}) e^{-c \min\{\theta^2/t^3, 1/t\}} + O(t^{-2} e^{(A-\eta)/t}). \quad (78)$$

Thus no other root of unity can contribute at algebraic order to the answer. This conclusion uses the entire circle, not merely a neighborhood of $q = 1$.

14 The discrete two variable saddle

Cauchy's formula gives

$$a(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \sum_{k \geq 1} T_k(e^{-t+i\theta}) e^{nt-in\theta} d\theta. \quad (79)$$

Choose $t = \sqrt{A/n}$ and $s = \sqrt{t}$. In the central region put

$$k = \frac{x_0}{t} + \frac{y}{s}, \quad \theta = t^{3/2}v, \quad z = t - i\theta = s^2(1 - isv), \quad x = kz = (x_0 + sy)(1 - isv). \quad (80)$$

The possible y form a lattice of mesh s , with a shift depending on t .

14.1 Finite Euler Maclaurin expansion

With B_{2r} denoting Bernoulli numbers, define

$$e_{2r-1}(x) = \frac{B_{2r}}{(2r)!} (f^{(2r-1)}(2x) - f^{(2r-1)}(x)). \quad (81)$$

Euler–Maclaurin applied to the finite sum $\sum_{j=k}^{2k} f(jz)$ gives, for every fixed $p \geq 1$,

$$T_k(e^{-z}) = e^{h(x)/z} g(x) \exp \left(\sum_{r=1}^p e_{2r-1}(x) z^{2r-1} + O(t^{2p-1}) \right). \quad (82)$$

The error estimate as written need not improve on the last displayed term; to obtain any specified finite power, apply this formula at a higher p and discard unnecessary terms. Uniformity holds when x and the line segment from x to $2x$ remain in fixed compact analytic neighborhoods of their positive real counterparts and $|z| = O(t)$.

Here are the remainder details. Apply the real-variable Euler–Maclaurin identity to the complex-valued function $u \mapsto f(uz)$ on $[k, 2k]$. Its $2p$ th derivative is $z^{2p} f^{(2p)}(uz)$. The periodic Bernoulli remainder is bounded by a constant times $k|z|^{2p} = O(t^{2p-1})$. The integral term is $h(kz)/z$ and the endpoint term is $(f(x) + f(2x))/2$. Combining the latter with e^{-3x} gives $g(x)$. Bounded derivatives on the compact neighborhoods establish the asserted uniformity.

14.2 Localization with explicit power bookkeeping

For the fixed ε, R above, set

$$c_h = \min_{x \in [\varepsilon, R]} \frac{A - h(x)}{(x - x_0)^2} > 0,$$

where the quotient at x_0 is defined as $a/2$. Let $L_t = \sqrt{K \log(1/t)}$, with K a sufficiently large fixed constant depending on the target order. The compact- k part with $|y| > L_t$ is bounded radially by $O(t^{-1} e^{A/t} t^{c_h K})$. In the full Cauchy integral, divided by its eventual leading scale $t e^{2A/t}$, this is $O(t^{c_h K - 2})$.

For $|v| > L_t$, equation (78) and $F(e^{-t}) = O(t^{-2} e^{A/t})$ give a relative bound $O(t^{c_h K - 3})$ plus an exponentially small term. For small enough t , $K \log(1/t) < 1/t$, as needed at the boundary of this region. The noncompact k tails are exponentially small by Lemma 12.2. Thus a relative remainder $O(t^N)$ permits restriction to

$$|y|, |v| \leq L_t \quad (83)$$

by taking $c_h K \geq N + 2$ and $c_h K \geq N + 3$. We may increase K further to cover the finitely many polynomial-Gaussian tails below. No choice here depends on n apart from the stated logarithmic window.

14.3 A uniform Gaussian remainder

After multiplying the summand by e^{nz} and extracting $g(x_0) e^{2A/t}$, its quadratic exponent is

$$Q(y, v) = -\frac{a}{2} y^2 + i a x_0 y v - \frac{D}{2} v^2. \quad (84)$$

For example, $A/z + nz - 2A/t = -Av^2 + O(sv^3)$, and the quadratic contribution of $h(x) - A$ is $-a(y - ix_0 v)^2/2$. This proves (84), with the D in (74). In particular $\Re Q$ is negative definite on the real (y, v) plane.

Lemma 14.1 (Uniform finite Taylor estimate). *Fix integers $J \geq 0$ and a fixed $K > 0$. There are polynomials $P_0, \dots, P_J$ in (y, v) , with $P_0 = 1$, an integer b_J , and positive constants C_J, t_J such that on (83), for $0 < t < t_J$,*

$$\left| \frac{T_k(e^{-z})e^{nz}}{g(x_0)e^{2A/t}} - e^{Q(y,v)} \sum_{m=0}^J P_m(y,v)s^m \right| \leq C_J s^{J+1} (1 + |y| + |v|)^{b_J} e^{-ay^2/4 - Dv^2/4}. \quad (85)$$

Furthermore $P_m(-y, -v) = (-1)^m P_m(y, v)$.

Proof. Use (82) at depth p such that $4p - 2 \geq J + 1$. Substitute (80) into its logarithm, subtract $2A/s^2 + \log g(x_0) + Q$, and denote the analytic part that remains by $R(s; y, v)$. The singular s^{-2} and s^{-1} terms cancel, so R is analytic in s at zero and $R(0; y, v) = 0$. Each of its Taylor coefficients is a polynomial in y, v ; the first has degree at most three.

Write $W = 1 + |y| + |v|$. For real $0 \leq \sigma \leq s$, with sW sufficiently small, the denominator $1 - i\sigma v$ stays bounded away from zero and $(x_0 + \sigma y)(1 - i\sigma v)$ stays in the same analytic neighborhood. Taylor's formula and the bounded derivatives of h, g , and the finitely many e_{2r-1} show that every fixed σ derivative of R is bounded by a fixed polynomial in W . They also give

$$|R(\sigma; y, v)| \leq C\sigma(W^3 + W).$$

To see the Gaussian domination explicitly, $W^3 \leq C_1 W(1 + y^2 + v^2)$, and $sW \rightarrow 0$ uniformly on (83). Hence, after reducing t_J ,

$$\Re R(\sigma; y, v) \leq \frac{a}{4} y^2 + \frac{D}{4} v^2 + C_2.$$

The derivatives of e^R are e^R times polynomials in fixed derivatives of R . Taylor's integral remainder for e^R at degree J , multiplied by e^Q , therefore gives (85). The Euler–Maclaurin error is $O(s^{4p-2})$ uniformly and is absorbed by the same majorant, using $|e^u - 1| \leq |u|e^{|u|}$.

Finally the substitution is invariant under $(s, y, v) \mapsto (-s, -y, -v)$; the remaining Euler–Maclaurin powers have the same invariance. The coefficient of s^m thus has the claimed parity. $\square$

The Gaussian estimate has no hidden logarithmic loss. Uniformly in the lattice shift and $0 < s \leq 1$,

$$s \sum_y \int_{\mathbb{R}} W^{b_J} e^{-ay^2/4 - Dv^2/4} dv \leq C'_J. \quad (86)$$

For example, split the real line into unit intervals, use the maximum of the polynomial times a slightly weaker Gaussian on each interval, and bound the number of lattice points there by $1 + s^{-1}$. The resulting summable Gaussian series proves (86). Thus the relative integrated remainder in (85) is $O(s^{J+1})$.

14.4 The lattice produces no algebraic oscillations

Each polynomial-Gaussian term can be extended to the full (y, v) plane with an error of arbitrarily high fixed order by the choice of K . Integrate in v first. Gaussian integration turns $P_m(y, v)e^{Q(y,v)}$ into a polynomial in y times

$$\exp \left[-\frac{1}{2} \left(a + \frac{a^2 x_0^2}{D} \right) y^2 \right] = \exp \left[-\frac{aA}{D} y^2 \right].$$

Poisson summation for a polynomial times this Gaussian replaces its shifted mesh- s sum by s^{-1} times its integral with exponentially small error. Indeed its Fourier transform is a polynomial times $\exp(-D\xi^2/(4aA))$. At nonzero frequencies $\xi = 2\pi m/s$, its absolute value is bounded by a polynomial in $|m|/s$ times

$$\exp\left(-\frac{\pi^2 D m^2}{a A t}\right), \quad m \neq 0. \quad (87)$$

The lattice shift only contributes unimodular phases. This also proves uniformity in the fractional part of x_0/t . Poisson summation here can equivalently be proved by Fourier expansion of the periodized Schwartz function; its absolutely convergent Fourier series permits termwise evaluation.

The substitution $(y, v) \mapsto (-y, -v)$ and Lemma 14.1 make all odd-degree full-plane integrals vanish. Take $J = 2M + 1$ to obtain an integer-power expansion in t , with relative error $O(t^{M+1})$. The leading term, including $d\theta = t^{3/2}dv$ and lattice density s^{-1} , is

$$\begin{aligned} \frac{g(x_0)te^{2A/t}}{2\pi} \int_{\mathbb{R}^2} e^{Q(y,v)} dy dv &= \frac{g(x_0)te^{2A/t}}{\sqrt{a(D + ax_0^2)}} \\ &= \frac{g(x_0)te^{2A/t}}{\sqrt{2aA}} = Cn^{-1/2}e^{B\sqrt{n}}. \end{aligned}$$

This proves the existence and remainder claim of Theorem 11.1. It remains to identify the coefficients.

15 The finite coefficient rule

Put $\ell(x) = \log g(x)$ and $H_m = h^{(m)}(x_0)$. Let Y be a formal centered Gaussian with variance $1/a$, meaning that the linear functional $\mathbb{E}$ on polynomials is determined by

$$\mathbb{E}Y^{2m} = (2m-1)!! a^{-m}, \quad \mathbb{E}Y^{2m+1} = 0, \quad \mathbb{E}1 = 1.$$

For every $j \geq 0$ define

$$\begin{aligned} c_j = [s^{2j}]\mathbb{E} \exp \left(\sum_{m \geq 3} \frac{H_m}{m!} Y^m s^{m-2} + \sum_{m \geq 1} \frac{\ell^{(m)}(x_0)}{m!} Y^m s^m \right. \\ \left. + \sum_{r \geq 1} \sum_{m \geq 0} \frac{e_{2r-1}^{(m)}(x_0)}{m!} Y^m s^{4r-2+m} \right). \end{aligned} \quad (88)$$

This is a finite rule: discard every monomial of s -degree greater than $2j$ before exponentiating, and truncate the exponential at the same degree. Thus one needs H_m only through $m = 2j + 2$, $\ell^{(m)}$ only through $2j$, and $4r - 2 + m \leq 2j$ in the last sum. All derivatives in this rule are algebraic numbers in $\mathbb{Q}(\sqrt{5})$; hence so is every c_j .

One way to check the algebra is to set $y = e^{-x}$ and differentiate using $-y d/dy$. For $m \geq 2$,

$$H_m = 2^m f^{(m-1)}(2x_0) - f^{(m-1)}(x_0),$$

while $\ell(x) = -3x + \frac{1}{2}f(x) + \frac{1}{2}f(2x)$. At x_0 the arguments are ϕ^{-1} and ϕ^{-2} , both in the same quadratic field. Some low-order checks are

$$\begin{aligned} H_2 &= 4 - 3\phi, & H_3 &= 7 - 2\phi, & H_4 &= 21 - 24\phi, \\ \ell'(x_0) &= -3\phi/2 - 2, & \ell''(x_0) &= \phi + 5/2, \\ e_1(x_0) &= 1/12, & e_3(x_0) &= -(\phi + 1)/120. \end{aligned}$$

Equation (88) is exactly the local real Laplace expansion: substitute $x = x_0 + \sqrt{t}Y$ in (82), divide by the leading Gaussian, and take its moments. The same localization and shifted-lattice estimates as in Section 14, with $v = 0$, show at every fixed order that

$$F(e^{-t}) = \lambda t^{-1/2} e^{A/t} \left(\sum_{j=0}^M c_j t^j + O_M(t^{M+1}) \right). \quad (89)$$

This supplies an elementary specialization of the prior radial theory; no originality is claimed for (89). Applying (88) gives (70). The package computes these values using exact rational pairs and checks them with a second finite polynomial calculation.

[write] **A check of the rule, 7 October 2026.** The write evaluated the finite rule (88) with its own code (mpmath at 50 digits, truncated power series in s with polynomial coefficients in Y , the Euler–Maclaurin terms through $r = 2$): $c_0, \dots, c_4$ agree with (70) to better than 10^{-43} , and the odd s -degrees vanish. It also confirmed the displayed values of $H_2, H_3, H_4, \ell'(x_0), \ell''(x_0), e_1(x_0), e_3(x_0), h(x_0) = A$ and λ , and $a = 3\phi - 4 = 0.8541019662\dots$, $D = 0.4601936581\dots > 0$. This is a numerical check of the stated algebra, not a second derivation.

16 Identification with the coefficient saddle

We justify the relation between (88) and the actual coefficients without assuming a sectorial version of (89).

Lemma 16.1 (Finite Gaussian contour identity). *For a polynomial P and an affine line $\Gamma = b + e^{i\alpha}\mathbb{R}$, oriented in the increasing real parameter, with $|\alpha| < \pi/4$,*

$$\int_{\Gamma} P(u) e^{-au^2/2} du = \int_{\mathbb{R}} P(u) e^{-au^2/2} du.$$

Proof. First translate a truncated line and then rotate it within the sectors of Gaussian decay. The integrand is entire. Along the connecting segments at radius R , its modulus is bounded by a polynomial in R times $\exp(-cR^2 + O(R))$, for some $c > 0$ if the rotation remains in a closed subsector of $|\alpha| < \pi/4$. Their integrals tend to zero. Cauchy’s theorem and passage to the limit prove the identity. A fixed translation does not alter the quadratic decay. $\square$

All coefficient calculations below are finite Taylor and Gaussian moment calculations already justified by Lemma 14.1; the contour identity does not transfer an uncontrolled real-axis error. In (80) use the principal square root and set $u = (x - x_0)/\sqrt{z}$. As real y varies, its line is

$$u = y\sqrt{1 - isv} - \frac{ix_0v}{\sqrt{1 - isv}}, \quad dk = z^{-1/2} du. \quad (90)$$

For fixed real v and small s , its direction lies inside a Gaussian decay sector. At each finite Taylor order the amplitude is a polynomial in u . Lemma 16.1 therefore identifies the k -saddle moments with the radial moments (88), giving the finite formal expression

$$\lambda z^{-1/2} e^{A/z} \sum_j c_j z^j. \quad (91)$$

Odd moments in $\sqrt{z}$ vanish.

This coefficientwise argument is compatible with the remaining v integral. Before contour translation the absolute quadratic factor is $e^{-ay^2/2 - Dv^2/2}$. A translated-contour bound may introduce

$e^{ax_0^2 v^2/2}$, but it is accompanied by e^{-Av^2} from the outer saddle, leaving $e^{-Dv^2/2}$. Since $D > 0$, every resulting finite polynomial moment is absolutely integrable. The actual remainders still come from (85), not from any assertion about (91) away from a finite expansion.

For clarity, the remaining one-dimensional moment algebra can be encoded by the modified Bessel saddle. The j th term corresponds to

$$(A/n)^{j/2+1/4} I_{-j-1/2}(2\sqrt{An}). \quad (92)$$

This notation records the formal dominant-exponential saddle of the contour integrand $z^{j-1/2} e^{A/z+nz}$; it is not an equality for a separately truncated Cauchy arc. The leading Gaussian normalizes $I_\nu(X)$ by $e^X/\sqrt{2\pi X}$. For a direct check of every coefficient, write its dominant series as $e^X X^{-1/2} \sum_{m \geq 0} b_m X^{-m}$, $b_0 = 1$. Substitution into

$$X^2 I_\nu'' + X I_\nu' - (X^2 + \nu^2) I_\nu = 0$$

gives

$$b_m = -\frac{4\nu^2 - (2m-1)^2}{8m} b_{m-1}.$$

The same differential identity for the formal saddle follows by differentiating the contour integrand and integrating a total derivative; Gaussian tails make its boundary term zero at each finite order. Thus the recurrence determines its moments uniquely. For $\nu = -j - 1/2$ it terminates at $m = j$, since

$$\prod_{\ell=1}^m ((2j+1)^2 - (2\ell-1)^2) = 4^m \frac{(j+m)!}{(j-m)!} \quad (0 \leq m \leq j).$$

Accordingly

$$I_{-j-1/2}(X) \sim \frac{e^X}{\sqrt{2\pi X}} \sum_{m=0}^j \frac{(-1)^m (j+m)!}{m! (j-m)! (2X)^m} \quad (93)$$

at algebraic order relative to e^X ; the other exponential is negligible. These are also the standard coefficients in DLMF [15]. Combining (91)–(93), the term indexed by j, m has relative $n^{-(j+m)/2}$ coefficient

$$c_j A^{j/2} \frac{(-1)^m (j+m)!}{m! (j-m)! (4\sqrt{A})^m}.$$

Put $r = j + m$. The condition $0 \leq m \leq j$ is exactly $\lceil r/2 \rceil \leq j \leq r$, proving (69) and completing Theorem 11.1.

17 Inverse expansions and integer thresholds

Define the formal logarithmic coefficients

$$\ell_r = [z^r] \log \left(1 + \sum_{j \geq 1} d_j z^j \right), \quad \alpha_r = B^r \ell_r. \quad (94)$$

Only finitely many terms enter at a fixed order. These ℓ_r are constants, distinct from the function $\ell(x) = \log g(x)$ in Section 15. For a large real threshold Y write

$$T = \log \frac{Y}{CB}, \quad w = \log T. \quad (95)$$

The variable $s = B\sqrt{n}$ used only in this section differs from the small parameter $s = \sqrt{t}$ in the saddle proof. The normalization is chosen so that Theorem 11.1 gives

$$\log \frac{a(n)}{CB} = s - \log s + \sum_{r=1}^J \alpha_r s^{-r} + O_J(s^{-J-1}). \quad (96)$$

Theorem 17.1 (Continuous models and two ceiling bounds). *For fixed $J \geq 1$ define polynomials $P_j(w)$ successively by*

$$\begin{aligned} S_0(u) &= 1 + uw, \\ P_j(w) &= [u^j] \left\{ \log S_{j-1}(u) - \sum_{r=1}^j \alpha_r u^r S_{j-1}(u)^{-r} \right\}, \\ S_j(u) &= 1 + uw + \sum_{i=1}^j P_i(w) u^{i+1}. \end{aligned} \quad (97)$$

Set

$$X_J(Y) = B^{-2} \left(T + \log T + \sum_{j=1}^J \frac{P_j(\log T)}{T^j} \right)^2. \quad (98)$$

Then $\deg P_j \leq j$, with

$$P_1(w) = w - \alpha_1, \quad P_2(w) = -\frac{1}{2}w^2 + (1 + \alpha_1)w - \alpha_1 - \alpha_2.$$

Along the exact values $Y = a(n)$,

$$X_J(a(n)) = n + O_J \left(\frac{(1 + \log \log a(n))^{J+1}}{(\log a(n))^J} \right). \quad (99)$$

For $N(Y) = \min\{n \geq 1 : a(n) \geq Y\}$ there is a constant $K_J > 0$ such that, for all sufficiently large Y ,

$$\lceil X_J(Y) - \epsilon_J(Y) \rceil \leq N(Y) \leq \lceil X_J(Y) + \epsilon_J(Y) \rceil, \quad \epsilon_J(Y) = K_J \frac{(1 + \log T)^{J+1}}{T^J}. \quad (100)$$

Alternatively let $s_J(Y)$ be the large positive root of

$$s - \log s + \sum_{r=1}^J \alpha_r s^{-r} = T. \quad (101)$$

It is eventually unique, and $Z_J(Y) = s_J(Y)^2/B^2$ satisfies

$$Z_J(a(n)) = n + O_J(n^{-J/2}). \quad (102)$$

There is also a two-ceiling bracket centered at $Z_J(Y)$ with radius $K'_J T^{-J}$ for some $K'_J > 0$ and all sufficiently large Y .

Proof. Let $H_J(s) = s - \log s + \sum_{r=1}^J \alpha_r s^{-r}$. For large s , $H'_J(s) = 1 + O(1/s) > 1/2$. Since $H_J(s) \rightarrow \infty$, it has a unique inverse on a sufficiently large half-line. Substitute $s = TS$ with $u = T^{-1}$ and $S = 1 + uw + \sum_{j=1}^J P_j(w)u^{j+1}$. The equation $H_J(s) = T$ is equivalent to

$$\sum_{j=1}^J P_j(w)u^j = \log S - \sum_{r=1}^J \alpha_r u^r S^{-r},$$

up to the truncation error. The coefficient of u^j on the right does not depend on P_j , which first enters S at degree $j + 1$. Hence (97) cancels the residual successively. Formal power-series multiplication proves by induction that $\deg P_j \leq j$; more explicitly every term contributing at u^j has total w -degree at most j . Evaluation at $w = \log T$ gives the residual bound $O_J((1 + w)^{J+1}T^{-J-1})$. These bounds also follow from convergent Taylor series of $\log(1 + v)$ and $(1 + v)^{-r}$ for $v = O(w/T)$. The mean value theorem then gives

$$s_J(Y) = T + \log T + \sum_{j=1}^J P_j(\log T)T^{-j} + O_J((1 + \log T)^{J+1}T^{-J-1}). \quad (103)$$

The error in (96) changes $s_J(a(n))$ by $O_J(s^{-J-1})$; squaring changes the index by $O_J(s^{-J})$. This proves (102). Squaring (103) costs one factor of T , proving (99).

For the integer assertion, introduce the smooth log model

$$L_J(x) = \log C - \frac{1}{2} \log x + B\sqrt{x} + \sum_{r=1}^J \ell_r x^{-r/2}.$$

Its derivative is $B/(2\sqrt{x}) + O(x^{-1}) > 0$ for large x and $\log a(n) = L_J(n) + O_J(n^{-(J+1)/2})$ for integers n . Let $x_* = Z_J(Y)$; equation (101) is exactly $L_J(x_*) = \log Y$. In a fixed relative neighborhood of x_* , the derivative is bounded below by a positive constant times $x_*^{-1/2}$, while the integer error is bounded by a constant times $x_*^{-(J+1)/2}$. Choose $\delta = Kx_*^{-J/2}$ with K sufficiently large. The mean value theorem ensures that every integer in that neighborhood with $n < x_* - \delta$ has $a(n) < Y$, and that $n = \lceil x_* + \delta \rceil$ has $a(n) \geq Y$.

For completeness, these local comparisons imply the global threshold assertion. Section 18 proves eventual strict increase. The finitely many earlier values are all below Y when Y is large. A nearby integer below $x_* - \delta$ bounds all earlier integers by eventual monotonicity. Therefore

$$\lceil x_* - \delta \rceil \leq N(Y) \leq \lceil x_* + \delta \rceil.$$

There is no endpoint ambiguity: $n < \lceil x_* - \delta \rceil$ implies $n < x_* - \delta$, whether or not $x_* - \delta$ is integral. Since $x_* \asymp T^2$, this gives the radius $K'_J T^{-J}$ around Z_J . Equation (103) implies $|x_* - X_J(Y)| = O_J((1 + \log T)^{J+1}T^{-J})$; enlarging its constant gives (100). $\square$

If the two ceilings in a bracket agree, their common value is the exact threshold. This is an asymptotic existence statement, not a certified numerical interval until its constants and range are made effective. In particular one may not infer $N(Y) = \lceil X_J(Y) \rceil$ for every large real Y : a threshold can be arbitrarily close to a jump. Even an $o(1)$ continuous error does not remove the need for rounding bounds. These inversion and ceiling mechanisms are standard asymptotic tools, also used in prior ProveIt reports [22]; no novelty is claimed for the general mechanism.

Remark 17.2 ([write] The inverses and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* For Y beyond the early values, $N(Y)$ is the integer staircase of `p0:def:three-inverses` with $A_n = a(n)$ and n_1 the onset of Corollary 18.1, so `p0:thm:staircase(1)` applies. In the b-file, the last n with $a(n+1) \leq a(n)$ is $n = 42$; that is a finite observation.
- (b) *Formal instance.* The construction (97) is `p0:thm:flattening(4)` with $a_{\text{vol}} = 1$, $b_{\text{vol}} = -1$, $q_r = \alpha_r$, $L_{\text{vol}} = T$ and $H_{\text{vol}} = \log T = w$: the volume’s ansatz $x = L - bH + \sum_n P_n(H)L^{-n}$ is the bracket in (98), and its Bell recurrence `p0:eq:P-bell` gives $P_1 = H - \alpha_1$ and $P_2 = -\frac{1}{2}H^2 + (1 + \alpha_1)H - \alpha_1 - \alpha_2$, the source’s P_1, P_2 (checked by hand and symbolically); the leading coefficient $[H^n]P_n = b^{n+1}/n$ of `p0:eq:P-shape` matches $-\frac{1}{2}$ at $n = 2$. The remainder in (103) is the source’s own; its residual-over-slope step is `p0:thm:backward-error`.

- (c) *Instance, not used by the source.* $H_J(s)$ is exactly of exponential–power type in s with data $(1, 1, 1, -1, 1, \sum_{r \leq J} \alpha_r \tau^r)$ (`p0:def:model`), so its large root $s_J(Y)$ also has the Lambert-centred expansion of `p0:thm:lambert-centered(5)` about $-W_{-1}(-e^{-T})$; the source expands about T instead, which is the flattened form of (b).
- (d) *Analogues.* The bracket (100) and the bracket about $Z_J(Y)$ are, as stated and proved, analogues of `p0:thm:staircase(2)`.

No novelty is claimed for any of these inversions; the source says the same of its mechanism.

18 Eventual increase and strict log concavity

Corollary 18.1. *The sequence $a(n)$ is eventually strictly increasing and eventually strictly log-concave. In particular, eventually $a(n)^2 > a(n-1)a(n+1)$.*

Proof. Retain the leading term and d_1 in (68); its relative remainder is $O(n^{-1})$. Taking the ratio at adjacent indices gives

$$\frac{a(n+1)}{a(n)} = 1 + \frac{B}{2\sqrt{n}} + O(n^{-1}) > 1.$$

No smoothness of that remainder is needed: the ratio of the two remainder factors is $1 + O(n^{-1})$. For log-concavity use the expansion through d_3 and take a logarithm:

$$\log a(n) = \log C - \frac{1}{2} \log n + B\sqrt{n} + \sum_{r=1}^3 \ell_r n^{-r/2} + O(n^{-2}).$$

The second difference of the remainder is still $O(n^{-2})$ by the triangle inequality. For the displayed smooth functions, Taylor’s theorem on $[n-1, n+1]$ gives

$$\log a(n+1) - 2\log a(n) + \log a(n-1) = -\frac{B}{4n^{3/2}} + O(n^{-2}) < 0.$$

The main negative term dominates. This proof neither differentiates nor assumes differentiability of an asymptotic error attached to a discrete sequence. $\square$

The argument proves no effective last exception. Finite searches are useful checks but cannot by themselves establish that their last observed failure is the global last failure.

19 Computation and reproducibility

The accompanying archive contains this TeX source, a compiled PDF, original exact code, generated tables, a manifest, and offline rebuild instructions. It does not include third-party papers, downloaded OEIS data, or private working reviews. Exact computations use integer and rational arithmetic in $\mathbb{Q}(\sqrt{5})$, with no conversion to floating point. Input and consistency guards use explicit exceptions so they remain active under `python -O`. The fixed-order generator accepts a nonnegative integer order and truncates by the entire degree of the expansion parameter; it is not restricted to the four displayed coefficients.

The exact checks use two independently organized finite Gaussian calculations, and partition counts are obtained from (66) by integer dynamic programming. A second small-range count routine checks the defining partition condition independently. The numerical diagnostics, when requested, use the separately disclosed optional `mpmath` dependency; they are not part of the exact proof. No network access is required by the rebuild.

For orientation, the relative errors of the coefficient approximations are defined as approximation divided by exact count, minus one. At $n = 10000$, the approximation through d_3 has error approximately $-3.3063 \cdot 10^{-7}$; including d_4 changes it to approximately $-1.4600 \cdot 10^{-8}$. The three-term implicit logarithmic inverse at the exact $a(10000)$ returns an index about 0.0000271 above 10000. These are diagnostics at selected arguments, not certified remainder bounds, optimal truncation claims, or evidence about exact integer rounding at arbitrary thresholds.

The source and generated tables make the calculation auditable without making the proof depend on numerical agreement. Section 14 proves the asymptotic remainder, while Section 15 and equation (69) specify the constants.

[write] The diagnostics recomputed, 7 October 2026. From (70) and (71), and again from the transfer formula (69), the write finds $d_1 = 2.424917491063005\dots$, $d_2 = 3.187616630712046\dots$, $d_3 = 9.439637377883243\dots$, $d_4 = 32.379308690155\dots$, whose twelve-place truncations are the values printed in Theorem 11.1. With $a(10000)$ from the OEIS b-file, the relative errors through d_3 and d_4 are $-3.30626\dots \cdot 10^{-7}$ and $-1.45998\dots \cdot 10^{-8}$, and the three-term implicit inverse returns $10000 + 2.709\dots \cdot 10^{-5}$, matching the approximate values printed above. In the b-file, the last $n \leq 9999$ with $a(n)^2 \leq a(n-1)a(n+1)$ is $n = 445$ and the last with $a(n+1) \leq a(n)$ is $n = 42$; finite observations, which prove no last exception.

20 Source comparison and limitations

The comparison below is deliberately bounded. It supports attribution and distinguishes neighboring models; it is not a priority certificate.

1. The OEIS entry A118096 [4] gives (66) and a leading equivalent attributed to Václav Kotěšovec, dated 13 June 2025. The absence of a proof beside an OEIS formula does not make it an open conjecture or establish novelty for a proof.
2. McIntosh's general-product result [18], in particular Theorem 2, and Zhou's Eulerian-series results [19], Theorems 1–2, supply prior real-axis all-order machinery. The present series is an Eulerian-series specialization, since

$$F(q) = q^3 \sum_{m \geq 0} q^{3m} \frac{(q^{2m+3}; q)_\infty}{(q^{m+1}; q)_\infty}.$$

The radial expansion and golden-ratio saddle are therefore prior-art territory. The complete-circle proof here is included to make the fixed-order coefficient argument self-contained, not to assert that such an argument is absent elsewhere.

3. Pittel [20] studies partitions with bounded largest-to-smallest ratio. Its publisher abstract identifies the condition $\max / \min \leq k$, $k > 1$, and a parameter satisfying $w^k + w = 1$. This is directly adjacent literature. The full theorem text was not accessible in the present comparison, so its enumerative precision and possible implications for this exact-boundary model remain unresolved here. No source clearance or exclusion of overlap is claimed.
4. The neighboring mock-theta models are distinct. Andrews, Dixit, and Yee [21] identify $q\chi_1(q)$ with $\max < 2 \min$ partitions and associate $\chi_0(q)$ with $\max \leq 2 \min$ and a unique smallest part. If

$$H(q) = \sum_{k \geq 1} \frac{q^k}{\prod_{j=k}^{2k} (1 - q^j)},$$

then direct termwise subtraction gives $F(q) = H(q) - q\chi_1(q)$. Indeed the strict class has summand $q^k / \prod_{j=k}^{2k-1} (1 - q^j)$; subtracting from the weak class leaves $q^{3k} / \prod_{j=k}^{2k} (1 - q^j)$. This does not identify H with a classical product or mock theta function, and the unique-minimum condition must not be dropped.

5. The Bessel coefficient product in Section 16 is classical [15]. Formal series reversion and derivative-based inverse control are standard; related inverse reports in ProveIt [22] are not a basis for claiming a new inversion method here.

The appropriate description is an independently derived, explicitly computable refinement with an incomplete historical comparison. No claim is made to have solved a previously unsolved problem.

21 Further questions

Several questions remain outside the proved scope.

- An accessible full reading of Pittel’s paper is needed to complete the closest historical comparison, before any originality assessment.
- Effective Euler–Maclaurin, localization, and Gaussian constants could turn the asymptotic inverse brackets into certified computable bounds and prove explicit last exceptions for increase or log-concavity.
- For largest part m times smallest part, $m \geq 3$, the finite interval becomes $[k, mk]$. The same architecture suggests further coefficient rules, but the constants, angular estimates, and inverse conclusions require their own proofs.
- The series $H(q)$ in Section 20 may admit useful exact identities. The elementary mock-theta subtraction alone does not settle its analytic structure.
- The present theorem fixes the order before $n \rightarrow \infty$. Growth of c_j , optimal truncation, and exponentially small corrections require analysis beyond the finite-order estimates proved here.

[write] 7 October 2026. Under Vladimir’s standing rule of 4 October 2026, the write found no claim of this Part that is unproved as stated and none that is wrong. The comparison with Pittel’s 2007 paper [20] remains open, as the first question says: the write’s attempt to read it at the publisher on 7 October 2026 was refused (HTTP 403), so neither the source nor the write has read its theorems, and nothing here should be read as excluding an overlap. By the source’s account (item 3 of Section 20), the abstract names a parameter with $w^k + w = 1$; at $k = 2$ its root in $(0, 1)$ is $\phi^{-1} = e^{-x_0}$, this Part’s saddle point, which makes the paper the closest comparison; that coincidence is all the write can add. On the second question, note only that in the b-file the last n with $a(n+1) \leq a(n)$ is $n = 42$ and the last with $a(n)^2 \leq a(n-1)a(n+1)$ is recorded in the note at the end of Section 19; finite searches prove no last exception.

Part III

Deficit asymptotics for partitions with divisible extreme parts

Bundle Report 218, archive `Report218-reproducibility.zip`, delivered as `report218.tex` (22 pages). Delivered title block “Deficit asymptotics for partitions with divisible extreme parts / Report 218 / 4 October 2026”. Printed as delivered, with [write] remarks and notes; Report 218’s Section k is Section $k + 21$ here, its equation (k) is equation $(k + 103)$, and its appendix keeps the letter A; statements move with their sections. Its labels carry `xpc:dv:`; its two unlabelled sections were labelled; its citation key `oeis` is `oeisA117086` and `nr` is `ngorhoades`. Its four generated tables are printed inline. This Part is self-contained.

Abstract of this Part (as delivered). Let $a(n)$ count partitions of n whose largest part is divisible by their smallest part, the sequence A117086, and put $D(n) = p(n) - a(n)$. We prove

$$D(n) \sim \frac{\pi}{2\sqrt{6n}} p(n)$$

and give its expansion to every fixed order, including an exact coefficient algorithm and six explicit relative coefficients. A finite-polynomial reduction of the fixed-minimum Fourier series and an elementary bound on the whole coefficient circle control the residue obstruction. The minimum tail is estimated before coefficient extraction; a finite-shift argument based on the Rademacher formula then supplies all orders. We also prove fixed-minimum residue equidistribution, a comparison with a reciprocal-minimum weighted count, sharpened to an explicit stretched-exponential bound for $n \geq 90$, finite- n enclosures, and all-order Lambert-branch inverse brackets. Exact rational code regenerates the coefficients, counting checks, and a certified enclosure at $n = 1000$. The elementary equivalence $a(n) \sim p(n)$ is background. No historical-priority claim, uniform growing-order expansion, or convergence claim is made.

22 The sequence and the main result

A partition is a finite nonincreasing list of positive integers. For a nonempty partition λ , write $|\lambda|$ for its sum and $\min \lambda$, $\max \lambda$ for its extreme parts. Let

$$a(n) = \#\{\lambda \vdash n : \min \lambda \mid \max \lambda\}, \quad D(n) = p(n) - a(n) \quad (n \geq 1),$$

where $p(n)$ is the unrestricted partition number and $p(0) = 1$. For computational bookkeeping only, set $a(0) = 0$. Thus $D(0)$ is not included in any generating series below.

The retrieved OEIS entry A117086, credited to Vladeta Jovovic on 17 April 2006, gives this definition and the generating function [5]

$$\sum_{n \geq 1} a(n) q^n = \sum_{L \geq 0} \sum_{k \geq 1} \frac{q^{(L+1)k}}{\prod_{i=k}^{Lk} (1 - q^i)}. \quad (104)$$

An empty product is one. The $L = 0$ term counts the one-part partition (k). For $L \geq 1$, removing one smallest part k and one largest part Lk leaves arbitrary parts between those endpoints. When $L = 1$, this counts at least two copies of k , so the singleton is counted exactly once overall.

Adjoining a part 1 is an injection from partitions of $n - 1$ to the partitions counted by $a(n)$. Consequently

$$p(n - 1) \leq a(n) \leq p(n) \quad (n \geq 1). \quad (105)$$

In particular, a is weakly increasing, since $a(n + 1) \geq p(n) \geq a(n)$. The familiar partition asymptotic already gives $a(n) \sim p(n)$. The quantity requiring further analysis is the deficit $D(n)$.

Throughout put

$$A = \frac{\pi^2}{6}, \quad N = n - \frac{1}{24}, \quad t = \sqrt{\frac{A}{N}}, \quad \mathcal{B}(n) = \frac{\exp(2\sqrt{AN})}{4\sqrt{3}N}. \quad (106)$$

Write $(q; q)_j = \prod_{h=1}^j (1 - q^h)$, $(q; q)_0 = 1$, and $P(q) = (q; q)_\infty^{-1} = \sum_{n \geq 0} p(n) q^n$. For the fixed-order expansions and fixed-minimum residue results, every order, minimum, modulus, and truncation depth is fixed before $n \rightarrow \infty$. Section 29 separately uses exact uniform inequalities at a specified growing auxiliary cutoff.

Theorem 22.1 (All fixed orders). *There are explicit polynomials $c_m \in \mathbb{Q}[A^{-1}]$, with $c_0 = 1$, such that for every integer $R \geq 0$,*

$$D(n) = \frac{\mathcal{B}(n)t}{2} \left(\sum_{m=0}^R c_m t^m + O_R(t^{R+1}) \right). \quad (107)$$

They are generated by (135) below. In particular,

$$\frac{D(n)}{p(n)} \sim \frac{\pi}{2\sqrt{6n}}.$$

The other principal conclusions are Theorem 25.2, concerning fixed-minimum residue classes; Corollary 28.1 and Theorem 29.1, comparing D with a rational weighted count; Theorem 26.1, an effective finite- n inequality; and Theorem 30.1, an all-order asymptotic inverse bracket. All proofs are given below, using the classical eta transformation and Rademacher partition formula as stated explicitly. These standard input identities are distinguished from the elementary estimates derived here.

Remark 22.2 ([write] The OEIS entry, quoted, 7 October 2026). OEIS A117086, read on 7 October 2026 in the internal format, is at revision #19 (3 September 2017), author Vladeta Jovovic (17 April 2006), offset 1. Its name is “Number of partitions of n such that the largest part is a multiple of the smallest part.”; its formula line is the generating function (104); a comment of Emeric Deutsch (21 April 2006) gives the conjugate reading (the number of parts is a multiple of the multiplicity of the largest part); the example “a(7)=12” lists the three excluded partitions [5,2], [4,3], [3,2,2]; it cross-references A118096 (Part II) and carries a b-file by Alois P. Heinz for $1 \leq n \leq 1000$. It states no asymptotic formula and no conjecture. The bibliography entry [5] prints the name as a title in its own words (“is divisible by” for “is a multiple of”). The revision number was visible to the write; like the source, it did not inspect the revision history. All 1000 b-file terms agree with the shipped table `data/218-divisible-generated-exact_values.csv`, whose partition numbers the write also recomputed, and brute-force counts agree for $n \leq 40$. Nothing was submitted to the OEIS.

23 Prior frameworks and source boundaries

The coefficient and residue methods used here have substantial antecedents. Bridges, Franke, and Garnowski [23] study the coefficients of $(\zeta q; q)_{\infty}^{-1}$ and residue biases for the number of parts. Partition conjugation gives the same unrestricted distribution for the largest part. Their Theorem 3.2 resolves much finer angular asymptotics than the whole-circle upper bound used in this report. Their Theorem 1.3 concerns the distinguished root $\zeta_b = \exp(2\pi i/b)$ for $b \geq 5$; it must not be read as the same formula for every primitive b th root. The angular classification excludes two crossover angles without proving they are irrational multiples of π . Our elementary bound applies to every nontrivial unit-modulus twist, so it does not need to resolve those crossover cases. A fixed minimum is an additional restriction, handled here by an exact polynomial reduction.

Ngo and Rhoades [11] prove arbitrary-order asymptotics for smallest-part parity and expected smallest-part size in Theorems 1.1 and 1.4. They credit their polynomial coefficient-transfer proposition, Proposition 1.8, to Wright. Cesana, Craig, and Males [24], Proposition 2.3 and Theorem 3.1, give general coefficient-transfer and equidistribution frameworks. Those transfer theorems require control away from the positive radius as well as a major-arc expansion. A relative little- o estimate on a twist alone does not establish agreement to every algebraic order. Here we prove an exponential Fourier bound and control the entire remaining minimum tail before extracting coefficients.

The weighted sum-of-tails identity in Section 28 is a specialization of established arbitrary-weight machinery. Andrews and Pedro Freitas [25] give the general Abel-lemma framework. We verified their publication metadata but did not inspect the original subscription-only full text. Gupta [26] reproduces their Theorem 4.1 as equation (1.9) in arXiv version 1 and equation (1.7) in the

published article. We give a direct absolute-convergence proof of the specialization used here; that proof does not close the original-text access gap.

The retrieved A117086 entry displays no asymptotic formula, but its live revision history was not accessible. A bounded literature screen did not locate the exact deficit theorem or reciprocal-minimum specialization presented here. Non-discovery does not establish novelty. This report makes no global novelty claim, and no quantum-modularity claim for the weighted series. The general twisted-product, all-order transfer, and arbitrary-weight tail frameworks are prior work.

24 Fixed-minimum Fourier reduction

For an integer $k \geq 1$, $|q| < 1$, and $|z| \leq 1$, define

$$H_k(z, q) = \sum_{\min \lambda = k} z^{\max \lambda} q^{|\lambda|}, \quad P_z(q) = \prod_{j \geq 1} (1 - zq^j)^{-1}.$$

The products and sums converge absolutely in this domain.

Lemma 24.1 (Exact polynomial reduction). *One has*

$$H_k(z, q) = q^k(q; q)_{k-1} P_z(q) + R_k(z, q), \quad (108)$$

$$R_k(z, q) = (zq)^k - q^k \sum_{M=0}^{k-1} (zq)^M \prod_{j=M+1}^{k-1} (1 - q^j). \quad (109)$$

The correction R_k is a polynomial of q -degree at most $k(k+1)/2$, and $R_k(1, q) = 0$.

Proof. For maximum $M > k$, removal of one endpoint of each size leaves the contribution $z^M q^{k+M} / \prod_{j=k}^M (1 - q^j)$. For $M = k$, the same expression counts constant partitions of multiplicity at least two; adding $(zq)^k$ restores exactly the singleton. Thus

$$H_k(z, q) = q^k(q; q)_{k-1} \sum_{M \geq k} \frac{(zq)^M}{(q; q)_M} + (zq)^k.$$

Euler's identity $\sum_{M \geq 0} u^M / (q; q)_M = (u; q)_\infty^{-1}$ proves (108) after removing $M < k$. For completeness, the series on the left satisfies $(1 - u)F(u) = F(qu)$ and $F(0) = 1$; iterating this equation proves the product identity for $|u| < 1$. Each summand removed has q -degree

$$k + M + \sum_{j=M+1}^{k-1} j = \frac{k(k+1)}{2} - \frac{M(M-1)}{2}.$$

Finally $\sum_{M=0}^{k-1} q^M / (q; q)_M = 1 / (q; q)_{k-1}$, by finite telescoping, so $R_k(1, q) = 0$. □

For example $R_1(z, q) = q(z - 1)$: finite corrections cannot be discarded at every index.

Let $F_k(n)$ count partitions of n with minimum exactly k . The lemma gives

$$\sum_{n \geq 1} F_k(n) q^n = q^k(q; q)_{k-1} P(q). \quad (110)$$

For $m \geq 2$, $0 \leq b < m$, put

$$C_{k,m,b}(n) = \#\{\lambda \vdash n : \min \lambda = k, \max \lambda \equiv b \pmod{m}\}.$$

Writing $\zeta_m = \exp(2\pi i/m)$, the elementary root filter is

$$C_{k,m,b}(n) = \frac{1}{m} \sum_{h=0}^{m-1} \zeta_m^{-bh} [q^n] H_k(\zeta_m^h, q). \quad (111)$$

This is an exact identity, including small indices. Fixing the minimum under conjugation fixes the multiplicity of the largest part, rather than the minimum of the conjugate partition.

25 An exponential bound on the whole circle

Lemma 25.1 (First harmonic gap). *If $0 < r < 1$, $|q| = r$, and $|\zeta| = 1$, then*

$$|P_\zeta(q)| \leq P(r) \exp\left(-\frac{r^2(1 - \operatorname{Re} \zeta)}{1 - r^2}\right). \quad (112)$$

Proof. Write $q = re^{i\theta}$ and $\zeta = e^{i\phi}$. Absolute convergence of the logarithmic series gives

$$\log P(r) - \log |P_\zeta(q)| = \sum_{h \geq 1} \frac{1}{h} \sum_{j \geq 1} r^{jh} (1 - \cos(h\phi + jh\theta)).$$

All summands are nonnegative. Retaining $h = 1$ gives $r/(1 - r) - \operatorname{Re}(\zeta q/(1 - q))$. As q traverses $|q| = r$, the point $q/(1 - q)$ traverses a circle with center $r^2/(1 - r^2)$ and radius $r/(1 - r^2)$. Hence

$$\operatorname{Re} \frac{\zeta q}{1 - q} \leq \frac{r^2 \cos \phi + r}{1 - r^2}.$$

Subtracting and exponentiating proves the claim. $\square$

This bound is uniform in the argument of q . Combining it with Lemma 24.1 and Cauchy's coefficient inequality gives, for $n > k(k + 1)/2$,

$$|[q^n] H_k(\zeta, q)| \leq r^{-n} P(r) r^k \prod_{j < k} (1 + r^j) \exp\left(-\frac{r^2(1 - \operatorname{Re} \zeta)}{1 - r^2}\right). \quad (113)$$

The bound is valid at every $0 < r < 1$ and will also be used for finite- n certification.

The classical eta transformation on the positive radius is

$$P(e^{-t}) = \sqrt{\frac{t}{2\pi}} \exp\left(\frac{A}{t} - \frac{t}{24}\right) P(e^{-4\pi^2/t}). \quad (114)$$

It follows from the modular identity for the Dedekind eta function [14, §23.18, eqs. (23.18.5)–(23.18.7)]; it is one of the standard partition identities used here. Since $P(x) = 1 + O(x)$ as $x \rightarrow 0$, our choice of t yields

$$\frac{e^{nt} P(e^{-t})}{\mathcal{B}(n)} = \frac{4\sqrt{3}A}{\sqrt{2\pi}} t^{-3/2} \left(1 + O(e^{-4\pi^2/t})\right). \quad (115)$$

Also $r^2/(1 - r^2) = 1/(2t) + O(1)$ at $r = e^{-t}$. Thus any fixed nontrivial root of unity gives an exponential saving, even after multiplication or division by any fixed power of t .

Theorem 25.2 (Fixed-minimum residue equidistribution). *For fixed $k \geq 1$, $m \geq 2$, and $0 \leq b < m$,*

$$F_k(n) = (k - 1)! \mathcal{B}(n) t^{k-1} (1 + O_k(t)), \quad (116)$$

$$\frac{C_{k,m,b}(n)}{F_k(n)} = \frac{1}{m} + O_{k,m}(e^{-\gamma_m/t}), \quad \gamma_m = \frac{1 - \cos(2\pi/m)}{4} > 0. \quad (117)$$

Proof. The finite-shift transfer in Lemma 27.1 below applied to $q^k(q; q)_{k-1}$ proves (116), because its radial expansion begins with $(k-1)!z^{k-1}$. For each nonzero h in (111), the real gap is at least $1 - \cos(2\pi/m)$. Equations (113) and (115), divided by (116), leave a fixed power of t times $\exp(-(1 - \cos(2\pi/m))/(2t) + O(1))$. Decreasing the exponent constant to γ_m absorbs that power. Summing the fixed number of twists proves the result. $\square$

There is no assertion uniform in a growing minimum k or modulus m .

26 The minimum tail and a finite enclosure

For $K \geq 2$, define the finite polynomial

$$S_K(q) = \sum_{k=2}^K \frac{k-1}{k} q^k(q; q)_{k-1}. \quad (118)$$

The root filter with $m = k$, $b = 0$ gives the exact decomposition

$$\sum_{n \geq 1} D(n)q^n = P(q)S_K(q) + E_K(q) + T_K(q), \quad E_K(q) = - \sum_{k=2}^K \frac{1}{k} \sum_{h=1}^{k-1} H_k(\zeta_k^h, q), \quad (119)$$

where T_K counts the actual deficit partitions with minimum at least $K+1$. In particular T_K has nonnegative integer coefficients. The polynomial corrections remain included in E_K until its coefficients are bounded.

For $|q| = r$ positivity gives

$$|T_K(q)| \leq T_K(r) \leq (q; q)_K \big|_{q=r} P(r) = \prod_{j=1}^K (1 - r^j) P(r). \quad (120)$$

The majorant counts all partitions with no parts $1, \dots, K$, including an irrelevant empty partition. At $r = e^{-t}$, Cauchy's inequality and (115) therefore prove

$$[q^n]T_K(q) = O_K(\mathcal{B}(n)t^{K-3/2}). \quad (121)$$

This explicitly retains the entire Cauchy loss $t^{-3/2}$. A radial Taylor expansion alone would not justify the coefficient estimate.

For $n > K(K+1)/2$, the polynomial corrections in E_K vanish. The fixed finite set of nontrivial twists and (113) make $[q^n]E_K$ exponentially small relative to $\mathcal{B}(n)$. Taking $K = R+4$ for any fixed $R \geq 0$ in (119) gives

$$D(n) = [q^n]P(q)S_{R+4}(q) + O_R(\mathcal{B}(n)t^{R+2}), \quad (122)$$

since $K - 3/2 = R + 5/2 > R + 2$.

There is also the stronger coefficientwise bound

$$0 \leq [q^n]T_K(q) \leq u_K(n) := [q^n]P(q)(q; q)_K = O_K(\mathcal{B}(n)t^K). \quad (123)$$

The last equality follows from Lemma 27.1; its leading term is $K!\mathcal{B}(n)t^K$. The weaker whole-circle argument already suffices for (122).

Theorem 26.1 (Explicit finite- n enclosure). *Let $K \geq 2$, $n > K(K+1)/2$, $0 < r < 1$, and put*

$$s_K(n) = [q^n]P(q)S_K(q), \quad (124)$$

$$\mathcal{E}_K(n, r) = r^{-n}P(r) \sum_{k=2}^K \frac{r^k \prod_{j < k} (1 + r^j)}{k} \sum_{h=1}^{k-1} \exp\left(-\frac{r^2(1 - \cos(2\pi h/k))}{1 - r^2}\right). \quad (125)$$

Then

$$s_K(n) - \mathcal{E}_K(n, r) \leq D(n) \leq s_K(n) + u_K(n) + \mathcal{E}_K(n, r). \quad (126)$$

The quantities $s_K(n)$ and $u_K(n)$ are finite rational combinations of shifted partition numbers.

Proof. Bound $[q^n]E_K$ in absolute value by (113), and use the coefficientwise bounds $0 \leq [q^n]T_K \leq u_K(n)$ in (119). $\square$

Increasing K need not improve this elementary enclosure: the least Fourier gap becomes smaller. Section 31 gives an entirely rational evaluation, with no floating exponentials or cosines.

27 Finite shifts and exact correction coefficients

27.1 The classical partition input

The Rademacher series, in the normalization of Johansson [12, eqs. (1.4)–(1.8)], is

$$p(n) = \frac{1}{2\sqrt{3}N} \sum_{a \geq 1} \frac{A_a(n)}{\sqrt{a}} U\left(\frac{2\sqrt{a}N}{a}\right), \quad U(x) = \cosh x - \frac{\sinh x}{x}. \quad (127)$$

The usual Rademacher sum $A_a(n)$ is a sum of unit-modulus phases over reduced residue classes modulo a ; hence $A_1(n) = 1$ and $|A_a(n)| \leq a$. No sharper estimate is needed. Since

$$U(x) = \frac{1}{x} \int_0^x u \sinh u \, du \leq \frac{x^2 e^x}{3} \quad (x \geq 0),$$

the $a \geq 2$ part is $O(\exp(\sqrt{a}N))$: after division by N its absolute sum is bounded by a constant times $\exp(\sqrt{a}N) \sum_{a \geq 2} a^{-3/2}$. The growing exponential in the $a = 1$ term gives

$$p(n) = f(N) + O(e^{\sqrt{a}N}), \quad f(N) = \frac{e^{2\sqrt{a}N}}{4\sqrt{3}N} \left(1 - \frac{1}{2\sqrt{a}N}\right). \quad (128)$$

The decreasing exponential is absorbed in the same error. This estimate holds at every one of finitely many fixed integer shifts as $n \rightarrow \infty$.

27.2 A terminating derivative formula

For $\ell \geq 0$, define

$$Q_\ell(t) = \sum_{h=0}^{\ell+1} \frac{(-1)^h (\ell + h + 1)!}{(\ell + 1 - h)! h!} \left(\frac{t}{4A}\right)^h. \quad (129)$$

Direct differentiation gives the exact identity

$$f^{(\ell)}(N) = \mathcal{B}(n) t^\ell Q_\ell(t). \quad (130)$$

Indeed, $Q_0(t) = 1 - t/(2A)$, $dt/dN = -t^3/(2A)$, and

$$Q_{\ell+1}(t) = \left(1 - \frac{(\ell+2)t}{2A}\right) Q_\ell(t) - \frac{t^2}{2A} Q'_\ell(t). \quad (131)$$

The factorial expression (129) satisfies this recurrence, proving (130) by induction.

Lemma 27.1 (Finite-polynomial transfer). *For a fixed polynomial $H(q)$ and a fixed integer $L \geq 0$,*

$$[q^n]P(q)H(q) = \mathcal{B}(n) \sum_{\ell=0}^L [z^\ell]H(e^{-z}) t^\ell Q_\ell(t) + O_{H,L}(\mathcal{B}(n)t^{L+1}). \quad (132)$$

Proof. Write $H(q) = \sum_{j=0}^J h_j q^j$. The coefficient is $\sum_j h_j p(n-j)$. Replace each partition number by $f(N-j)$ using (128), with an exponentially small total error. Taylor-expand this explicit smooth function at N to degree L . By (130), its $(L+1)$ st derivative throughout each bounded interval $[N-J, N]$ is $O_{H,L}(\mathcal{B}(n)t^{L+1})$. The Taylor remainder has the stated size. Finally,

$$\sum_{j=0}^J h_j \frac{(-j)^\ell}{\ell!} = [z^\ell]H(e^{-z}).$$

Only the explicit function f is differentiated; no unknown error term on the integer sequence $p(n)$ is differentiated. $\square$

27.3 The formal coefficient algorithm

Define a formal power series

$$G(z) = \sum_{k \geq 2} \frac{k-1}{k} e^{-kz} \prod_{j=1}^{k-1} (1 - e^{-jz}), \quad g_\ell = [z^\ell]G(z). \quad (133)$$

The k th term has valuation $k-1$, so any specified coefficient involves finitely many rational operations. Finite telescoping using $q^k(q; q)_{k-1} = (q; q)_{k-1} - (q; q)_k$ proves the equivalent formal identity

$$G(z) = \sum_{j \geq 1} \frac{\prod_{h=1}^j (1 - e^{-hz})}{j(j+1)}. \quad (134)$$

No analytic convergence near $z = 0$ is asserted. Define

$$\sum_{m \geq 0} c_m t^m = \frac{2}{t} \sum_{\ell \geq 1} g_\ell t^\ell Q_\ell(t), \quad (135)$$

$$c_m = 2 \sum_{\substack{1 \leq \ell \leq m+1 \\ h=m+1-\ell \leq \ell+1}} g_\ell \frac{(-1)^h (\ell+h+1)!}{(\ell+1-h)! h! (4A)^h}. \quad (136)$$

The displayed sum always has $h \geq 0$. Each coefficient belongs to $\mathbb{Q}[A^{-1}]$ and is effectively computable.

Proof of Theorem 22.1. Use (122) with fixed R , then Lemma 27.1 with $H = S_{R+4}$ and $L = R+1$. The coefficients of $S_{R+4}(e^{-z})$ through degree $R+1$ equal g_ℓ , because every omitted minimum has larger valuation. Terms of t -degree above $R+1$ in the finite sum contribute $O_R(\mathcal{B}(n)t^{R+2})$. Equations (135)–(136) now give (107). The first radial coefficient is $g_1 = 1/2$, so $c_0 = 1$. Since $p(n) = \mathcal{B}(n)(1 - t/(2A)) + O(e^{\sqrt{AN}})$ and $t \sim \pi/\sqrt{6n}$, the leading equivalent follows. $\square$

The first eight radial coefficients are

$$(g_1, \dots, g_8) = \left(\frac{1}{2}, \frac{1}{12}, \frac{1}{12}, \frac{89}{720}, \frac{61}{240}, \frac{20287}{30240}, \frac{21841}{10080}, \frac{10023809}{1209600} \right).$$

The first six relative coefficients in (107) are

$$\begin{aligned} c_0 &= 1, \\ c_1 &= \frac{1}{6} - \frac{3}{2A}, \\ c_2 &= \frac{1}{6} - \frac{1}{2A} + \frac{3}{4A^2}, \\ c_3 &= \frac{89}{360} - \frac{5}{6A} + \frac{5}{8A^2}, \\ c_4 &= \frac{61}{120} - \frac{89}{48A} + \frac{15}{8A^2} - \frac{5}{16A^3}, \\ c_5 &= \frac{20287}{15120} - \frac{427}{80A} + \frac{623}{96A^2} - \frac{35}{16A^3}. \end{aligned}$$

Relative to the exact partition number, divide the formal series by $1 - t/(2A)$ to obtain

$$\begin{aligned} \frac{D(n)}{p(n)} &= \frac{t}{2} \left(1 + \rho_1 t + \rho_2 t^2 + \rho_3 t^3 + O(t^4) \right), \\ \rho_1 &= \frac{1}{6} - \frac{1}{A}, \\ \rho_2 &= \frac{1}{6} - \frac{5}{12A} + \frac{1}{4A^2}, \\ \rho_3 &= \frac{89}{360} - \frac{3}{4A} + \frac{5}{12A^2} + \frac{1}{8A^3}. \end{aligned} \tag{137}$$

The shift $N = n - 1/24$ remains part of these formulas; replacing t by $\pi/\sqrt{6n}$ without re-expanding changes higher coefficients.

[write] **Part I's counterpart, 7 October 2026.** This section is, for the same partition function, Part I's Section 4: (128) is Lemma 4.1 (both from Johansson's normalization of the Rademacher series, with the same bound on U), (129), (130) and (131) are Lemma 4.2 (the same polynomials Q_ℓ), and Lemma 27.1 is Proposition 4.3; the eta identity (114) is (32). It is kept because the later sections of this Part cite these statements by their own labels; the proofs are the same argument.

28 The reciprocal-minimum weighted count

Define

$$W(n) = \sum_{\lambda \vdash n} \left(1 - \frac{1}{\min \lambda} \right) \quad (n \geq 1).$$

This rational number is a weighted count, rather than an integer count of a subset.

Corollary 28.1. *For every fixed real $L > 0$,*

$$D(n) - W(n) = O_L(p(n)n^{-L}). \tag{138}$$

Proof. Choose a fixed integer $K \geq 2L + 2$. For minima $k \leq K$, the difference between the deficit count and the weight $(k - 1)/k$ times the fixed-minimum count is a finite sum of nontrivial Fourier coefficients. These are exponentially small relative to $p(n)$ by Section 25. Each remaining nonnegative tail, for D or W , is at most $u_K(n)$, because its weights lie in $[0, 1]$. Equation (123) bounds their sum by $O_K(p(n)n^{-K/2})$. This implies (138). $\square$

The minimum and its associated modulus range over all integers in the complete sum. This fixed-cutoff proof alone gives no specified decay rate beyond all algebraic orders. Section 29 strengthens it using exact uniform inequalities at a growing auxiliary cutoff.

For completeness, the exact weighted generating function is

$$\sum_{n \geq 1} W(n)q^n = P(q)H(q), \quad H(q) = \sum_{k \geq 2} \frac{k-1}{k} q^k (q; q)_{k-1}. \quad (139)$$

For $|q| < 1$, ordinary telescoping gives

$$H(q) = \sum_{j \geq 1} \frac{(q; q)_j - (q; q)_\infty}{j(j+1)} = \sum_{j \geq 1} \frac{(q; q)_j}{j(j+1)} - (q; q)_\infty. \quad (140)$$

The last product is indispensable in the analytic identity, even though it has no contribution to the formal radial series (134). Nor is $P(q)H(q)$ the exact deficit generating function: the Fourier terms still distinguish the two.

Here is a direct derivation of the connection to the arbitrary-weight tail framework mentioned in Section 23. Define

$$g(x) = \sum_{j \geq 1} \frac{x^j}{j(j+1)} = 1 + (x^{-1} - 1) \log(1 - x), \quad |x| < 1,$$

with the removable value $g(0) = 0$ and logarithm given by its power series. Euler's identity gives

$$(q; q)_j - (q; q)_\infty = (q; q)_\infty \sum_{v \geq 1} \frac{q^{v(j+1)}}{(q; q)_v}.$$

For $|q| \leq r < 1$, finite products are bounded above by $\prod_{h \geq 1} (1 + r^h)$ and their reciprocals by $P(r)$. The double series of absolute values is bounded by a constant times

$$\sum_{j \geq 1} \frac{1}{j(j+1)} \sum_{v \geq 1} r^{v(j+1)} < \infty.$$

Interchanging sums is therefore justified and yields

$$H(q) = (q; q)_\infty \sum_{v \geq 1} \frac{q^v g(q^v)}{(q; q)_v}. \quad (141)$$

Since $q^v g(q^v) = q^v + (1 - q^v) \log(1 - q^v)$, this also proves

$$P(q)H(q) = P(q) - 1 + \sum_{v \geq 1} \frac{\log(1 - q^v)}{(q; q)_{v-1}}. \quad (142)$$

These are exact analytic identities for the weighted statistic. They are not substitutes for the coefficient error estimates that prove Corollary 28.1.

29 A quantitative reciprocal-minimum approximation

The exact estimates proved earlier also permit a specified growing auxiliary cutoff. This yields a stronger global comparison than Corollary 28.1, without using a coefficient-transfer theorem uniform in that cutoff.

Theorem 29.1 (Explicit stretched-exponential comparison). *Put*

$$c_* = \left(\frac{\pi\sqrt{6}}{9} \right)^{1/3}, \quad C_0 = \frac{32\sqrt{3}A}{\sqrt{2\pi}}, \quad \Lambda = \log(N/A).$$

For every integer $n \geq 90$,

$$\frac{|D(n) - W(n)|}{p(n)} \leq C_0 \exp \left(-c_* N^{1/6} \Lambda^{2/3} + 3c_* \log 2 N^{1/6} \Lambda^{-1/3} + \frac{5}{4} \Lambda \right). \quad (143)$$

Consequently, with an absolute implied constant,

$$\frac{|D(n) - W(n)|}{p(n)} \leq \exp \left(-c_* n^{1/6} (\log n)^{2/3} + O(n^{1/6} (\log n)^{-1/3}) \right). \quad (144)$$

For each fixed $\varepsilon > 0$, the right side can therefore be replaced, for all sufficiently large n , by $\exp(-(c_* - \varepsilon)n^{1/6}(\log n)^{2/3})$.

The explicit bound may exceed one at small indices; it remains valid. It is an upper bound, not an asymptotic formula for the actual error. In particular it does not assert a rate $\exp(-c\sqrt{n})$ for any fixed $c > 0$.

29.1 An exact bound uniform in the cutoff

For any integer $K \geq 2$, $n > K(K+1)/2$, and $0 < r < 1$, define

$$\Gamma_K(r) = \frac{r^2(1 - \cos(2\pi/K))}{1 - r^2}.$$

Then the following bound is simultaneous in all its displayed parameters:

$$|D(n) - W(n)| \leq r^{-n} P(r) \left(2^K e^{-\Gamma_K(r)} + \prod_{j=1}^K (1 - r^j) \right). \quad (145)$$

To prove it, let $C_k(n) = C_{k,k,0}(n)$. The minimum- k contribution to $D - W$ is $F_k(n)/k - C_k(n)$. For $2 \leq k \leq K$, the root filter and polynomial reduction give the negative sum of the nontrivial twists, divided by k . Every polynomial coefficient has vanished under the stated bound on n . All nontrivial k th roots have $1 - \operatorname{Re} \zeta \geq 1 - \cos(2\pi/K)$, because their angular distance from 1 is at least $2\pi/k \geq 2\pi/K$. Hence (113) bounds the total small-minimum error by

$$r^{-n} P(r) e^{-\Gamma_K(r)} \sum_{k=2}^K \frac{k-1}{k} r^k \prod_{j < k} (1 + r^j) < r^{-n} P(r) 2^K e^{-\Gamma_K(r)}.$$

For a partition with minimum $k > K$, its signed weight in $D - W$ is $1/k - \mathbf{1}_{k|\max \lambda}$, whose absolute value is at most one. Thus the absolute tail is bounded by the restricted-partition coefficient $u_K(n)$ and then by its positive-radius Cauchy majorant. This proves (145), with only one tail majorant. At $r = e^{-t}$, it gives

$$|D(n) - W(n)| \leq e^{nt} P(e^{-t}) \left(2^K \exp \left(-\frac{1 - \cos(2\pi/K)}{e^{2t} - 1} \right) + t^K K! \right), \quad (146)$$

since $1 - e^{-jt} \leq jt$. No parameter-dependent asymptotic error occurs in this inequality.

29.2 An explicit normalized Cauchy cost

When $t \leq e^{-2}$, the exact eta identity (114) and the Rademacher series give

$$\frac{e^{nt}P(e^{-t})}{p(n)} \leq C_1 t^{-3/2}, \quad C_1 = \frac{16\sqrt{3}A}{\sqrt{2\pi}}. \quad (147)$$

Here are explicit bounds establishing the constant and its onset. Set $w = 2A/t$. The tail estimate for (127), using $\sum_{a \geq 2} a^{-3/2} \leq \int_1^\infty x^{-3/2} dx = 2$, gives

$$\frac{p(n)}{\mathcal{B}(n)} \geq 1 - \frac{1}{w} - \frac{4w^2}{3} e^{-w/2}. \quad (148)$$

The decreasing exponential in the $a = 1$ summand is positive and has been discarded. At $t \leq e^{-2}$, elementary rational bounds imply

$$w \geq 2Ae^2 > \frac{(25/8)^2(163/60)^2}{3} = \frac{664225}{27648} > 24.$$

Since $w^2 e^{-w/2}$ decreases for $w \geq 4$,

$$\frac{p(n)}{\mathcal{B}(n)} \geq 1 - \frac{1}{24} - 768e^{-12} > 1 - \frac{1}{24} - 768\left(\frac{3}{8}\right)^{12} > \frac{1}{2}.$$

For $0 < s \leq 1/10$,

$$\log P(s) \leq \sum_{j \geq 1} \frac{s^j}{1-s^j} \leq \frac{s}{(1-s)^2} \leq \frac{10}{81} < \frac{1}{2} < \log 2.$$

Taking $s = e^{-4\pi^2/t} < 1/10$ gives $P(s) < 2$. The last strict bound follows already from $4\pi^2/t > 36 > 3$ and $e^3 > (8/3)^3 > 10$. Combining these estimates with the exact identity

$$\frac{e^{nt}P(e^{-t})}{\mathcal{B}(n)} = \frac{4\sqrt{3}A}{\sqrt{2\pi}} t^{-3/2} P(e^{-4\pi^2/t})$$

proves (147).

The requirement $t \leq e^{-2}$ holds for every $n \geq 90$, because

$$Ae^4 + \frac{1}{24} < \frac{(22/7)^2(87/32)^4}{6} + \frac{1}{24} = \frac{6935272345}{77070336} < 90. \quad (149)$$

For clarity, the elementary constants used here admit short exact certificates. Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and alternating arctangent series give $25/8 < \pi < 22/7$, hence $9 < \pi^2 < 10$. Also

$$\frac{163}{60} = \sum_{j=0}^5 \frac{1}{j!} < e < \sum_{j=0}^6 \frac{1}{j!} + \frac{1/7!}{1-1/8} < \frac{87}{32}.$$

The upper tail bound follows because every subsequent ratio is at most $1/8$. These comparisons, including (149), are regenerated with rational arithmetic in the accompanying code. The inequality $\log 2 > 1/2$ follows directly from $\int_1^2 du/u > 1/2$.

29.3 Balancing a growing integer cutoff

Put

$$L = \log(1/t) \geq 2, \quad S = t^{-1/3}L^{2/3}, \quad \kappa = \left(\frac{3\pi^2}{2}\right)^{1/3}, \quad \beta_0 = \left(\frac{4\pi^2}{9}\right)^{1/3} = \frac{2\kappa}{3},$$

and choose

$$x = \kappa(tL)^{-1/3}, \quad K = \lfloor x \rfloor. \quad (150)$$

One has $x > 3$ and $tx < 1$. Indeed $x^3 = (3\pi^2/2)e^L/L$ increases for $L \geq 2$, whereas $(tx)^3 = (3\pi^2/2)e^{-2L}/L$ decreases. At $L = 2$, the first is greater than $48 > 27$, and the second is less than $(30/4)(3/8)^4 < 1$. Thus $K \geq 3$, $K < 1/t$, and

$$\frac{K(K+1)}{2} < \frac{t^{-2} + t^{-1}}{2} < t^{-2} = \frac{N}{A} < n. \quad (151)$$

All polynomial corrections therefore vanish even for this growing cutoff. This is a direct check of the exact inequality's domain, rather than a uniform extension of Lemma 27.1.

For the twist term, $K \leq x$ implies $1 - \cos(2\pi/K) \geq 1 - \cos(2\pi/x)$. Use

$$1 - \cos u \geq \frac{u^2}{2} - \frac{u^4}{24}, \quad \frac{1}{e^{2t} - 1} \geq \frac{1}{2t} - \frac{1}{2}.$$

The first follows, for example, from the alternating cosine series for the present $u = 2\pi/x < 2\pi/3$; the second follows from $\coth t \geq 1/t$, since $t \cosh t - \sinh t$ has nonnegative derivative and vanishes at zero. Both right sides are nonnegative here: $u^2 < 4\pi^2/9 < 12$ and $t < 1$. Multiplying and dropping the positive final term gives

$$\Gamma_K(e^{-t}) \geq \frac{\pi^2}{tx^2} - \delta(t) = \beta_0 S - \delta(t), \quad \delta(t) = \frac{\pi^2}{x^2} + \frac{\pi^4}{3tx^4}. \quad (152)$$

Hence

$$2^K e^{-\Gamma_K(e^{-t})} \leq \exp(-\beta_0 S + x \log 2 + \delta(t)). \quad (153)$$

There is no omitted loss from rounding down: that rounding improves the lower bound on the angular gap.

To bound δ , write

$$\frac{\delta(t)}{L} = \frac{\pi^2}{\kappa^2} e^{-2L/3} L^{-1/3} + \frac{\pi^4}{3\kappa^4} e^{-L/3} L^{1/3}.$$

Both summands decrease for $L \geq 2$; their logarithmic derivatives are $-2/3 - 1/(3L)$ and $-1/3 + 1/(3L)$. At $L = 2$, their cubes are respectively

$$\begin{aligned} \frac{2\pi^2}{9} e^{-4} &< \frac{20}{9} \left(\frac{3}{8}\right)^4 < \left(\frac{9}{25}\right)^3, \\ \frac{32\pi^4}{2187} e^{-2} &< \frac{3200}{2187} \left(\frac{3}{8}\right)^2 < \left(\frac{3}{5}\right)^3. \end{aligned}$$

Thus $\delta(t)/L < 9/25 + 3/5 = 24/25 < 1$, and $\delta(t) \leq L$.

For the minimum tail set

$$H = -\log(tx) = \frac{2}{3}L + \frac{1}{3}\log L - \log \kappa > 0.$$

Using $K! \leq K^K$, $K \leq x$, and $K \geq x - 1$ gives

$$\log(t^K K!) \leq K \log(tK) \leq -KH \leq -(x-1)H.$$

Since $L \geq 2 > 3\pi^2/16 = \kappa^3/8$, one has $H \geq (2/3)L - \log 2$; also $H = L - \log x \leq L$. It follows that

$$\log(t^K K!) \leq -\beta_0 S + x \log 2 + L. \quad (154)$$

This accounts explicitly for the possible floor-rounding loss, at most $H \leq L$.

Equations (153)–(154) and the normalized Cauchy cost give, for every $n \geq 90$,

$$\frac{|D(n) - W(n)|}{p(n)} \leq 2C_1 \exp\left(-\beta_0 S + \kappa \log 2 \frac{S}{L} + \frac{5}{2}L\right). \quad (155)$$

Now $L = \Lambda/2$ and

$$\beta_0 S = c_* N^{1/6} \Lambda^{2/3}, \quad \kappa \log 2 \frac{S}{L} = 3c_* \log 2 N^{1/6} \Lambda^{-1/3}.$$

Indeed $c_*^3 = \beta_0^3 \Lambda^{-1/2}/4 = \pi^2/(9\sqrt{\Lambda}) = \pi\sqrt{6}/9$. Since $C_0 = 2C_1$, this proves (143). Finally

$$N^{1/6} \Lambda^{2/3} = n^{1/6} (\log n)^{2/3} + O(n^{1/6} (\log n)^{-1/3}).$$

The other terms in (143), including $O(\log n)$ and $\log C_0$, fit in this error scale. This proves (144) and the eventual ε statement, completing the proof of Theorem 29.1.

For a chosen $\varepsilon > 0$ and a particular integer $n \geq 90$, comparing the logarithm of the right side of (143) with $-(c_* - \varepsilon)n^{1/6}(\log n)^{2/3}$ is a computable pointwise sufficient test. The first passing integer is not asserted to certify every larger integer or an optimal onset for the true error. Eventual validity follows from (144).

29.4 Optimality only for this two-term majorant

The balance above optimizes the leading constant of the specific majorant

$$M(t, K) = 2^K e^{-\Gamma \kappa (e^{-t})} + t^K K!.$$

If $K \sim a(tL)^{-1/3}$ for a fixed $a > 0$, elementary small-argument expansions and the logarithmic Stirling formula yield

$$\frac{\log(2^K e^{-\Gamma \kappa})}{S} \longrightarrow -\frac{\pi^2}{a^2}, \quad \frac{\log(t^K K!)}{S} \longrightarrow -\frac{2a}{3}.$$

For the second limit, use $\log(tK) = -(2/3)L - (1/3)\log L + \log a + o(1)$ and $\log K! = K \log K - K + O(\log K)$. Therefore

$$\frac{-\log M(t, K)}{S} \longrightarrow \min \left\{ \frac{\pi^2}{a^2}, \frac{2a}{3} \right\},$$

whose unique maximum is β_0 at $a = \kappa$.

This also bounds arbitrary admissible integer-cutoff sequences with $K \geq 2$. Write $a(t) = K(tL)^{1/3}$ and $R(t) = -\log M(t, K)/S$. Since each term of M is positive,

$$R(t) \leq a(t), \quad R(t) \leq \frac{\pi^2}{a(t)^2}.$$

The first uses $\log K! \geq 0$; the second uses $1 - \cos u \leq u^2/2$ and $e^{2t} - 1 \geq 2t$ and discards $K \log 2 \geq 0$. If $\limsup R > \beta_0$, a subsequence with $R \geq \beta_0 + \eta$ confines $a(t)$ to a compact subset of $(0, \infty)$. A convergent further subsequence contradicts the preceding fixed- a limit. Thus no cutoff sequence improves this leading constant within M ; the choice (150) attains it. The logarithm of the Cauchy cost is $O(L) = o(S)$ and does not alter the conclusion.

This does not optimize the actual error, the sharper product tail in (145), a more detailed Fourier sum, or another decomposition. No absence of cancellation or actual-error asymptotic is proved, and subleading cutoff terms have not been optimized.

30 All-order inverses and integer thresholds

30.1 The common formal reversion

For $X = D$ or $X = a$, use the weak threshold convention

$$\nu_X(y) = \min\{n \geq 1 : X(n) \geq y\}.$$

Set $w = 2\sqrt{A(n-1/24)}$. Theorem 22.1 gives, at every fixed order,

$$X(n) = Ce^w w^{-\beta} \left(1 + \sum_{j=1}^R d_j w^{-j} + O_R(w^{-R-1}) \right). \quad (156)$$

For the deficit,

$$C_D = \frac{A^2}{\sqrt{3}}, \quad \beta_D = 3, \quad d_j^{(D)} = (2A)^j c_j. \quad (157)$$

In particular,

$$d_1^{(D)} = \frac{A}{3} - 3, \quad d_2^{(D)} = \frac{2A^2}{3} - 2A + 3, \quad d_3^{(D)} = \frac{A(89A^2 - 300A + 225)}{45}.$$

For $a = p - D$,

$$C_a = \frac{A}{\sqrt{3}}, \quad \beta_a = 2, \quad d_1^{(a)} = -(A+1), \quad d_j^{(a)} = -A d_{j-1}^{(D)} \quad (j \geq 2). \quad (158)$$

Indeed $p/\mathcal{B} = 1 - 1/w$ to all algebraic orders, whereas $D/\mathcal{B} = (A/w)(1 + \sum_{j \geq 1} d_j^{(D)} w^{-j})$.

For the relevant C, β , define

$$w_0 = -\beta W_{-1} \left(-\frac{(C/y)^{1/\beta}}{\beta} \right). \quad (159)$$

For $y > C(e/\beta)^\beta$, this is the unique solution $w_0 > \beta$ of $Ce^{w_0} w_0^{-\beta} = y$. The lower real Lambert branch W_{-1} is essential; the principal branch describes the other side of the minimum.

With $u = 1/w_0$, seek a formal series $\Delta(u) = \sum_{j \geq 1} v_j u^j$ satisfying

$$\Delta - \beta \log(1 + u\Delta) + \log \left(1 + \sum_{j \geq 1} d_j \left(\frac{u}{1 + u\Delta} \right)^j \right) = 0. \quad (160)$$

At degree u^m , the unknown v_m occurs with coefficient one only in Δ ; elsewhere its occurrence gains at least one additional u . Therefore the solution is unique and triangularly computable. The first coefficients are

$$\begin{aligned} v_1 &= -d_1, \\ v_2 &= -\beta d_1 + \frac{d_1^2}{2} - d_2, \\ v_3 &= -\beta^2 d_1 + \left(\frac{\beta}{2} - 1\right) d_1^2 - \beta d_2 - \frac{d_1^3}{3} + d_1 d_2 - d_3. \end{aligned} \tag{161}$$

For an integer $M \geq 1$, retain the square in full and define

$$x_M(y) = \frac{1}{24} + \frac{1}{4A} \left(w_0 + \sum_{j=1}^M \frac{v_j}{w_0^j} \right)^2. \tag{162}$$

Theorem 30.1 (Asymptotic integer inverse bracket). *For $X = D$ or $X = a$ and every fixed $M \geq 1$, there are constants $B_M > 0$ and y_M such that*

$$\left\lceil x_M(y) - B_M w_0^{-M} \right\rceil \leq \nu_X(y) \leq \left\lceil x_M(y) + B_M w_0^{-M} \right\rceil \quad (y \geq y_M). \tag{163}$$

The constants and onset are not made effective here.

Proof. Use the truncated smooth forward function

$$F_M(x) = C e^{w w^{-\beta}} \left(1 + \sum_{j=1}^M d_j w^{-j} \right), \quad w = 2\sqrt{A(x - 1/24)}.$$

It is positive and increasing for sufficiently large x . Its logarithmic derivative is

$$\frac{d}{dx} \log F_M(x) = \frac{2A}{w} \left(1 - \frac{\beta}{w} + O(w^{-2}) \right).$$

Substituting the truncated Δ into the logarithmic equation gives residual $O(w_0^{-M-1})$. The derivative of that equation with respect to w is $1 + O(w_0^{-1})$. Thus the error in the w root is $O(w_0^{-M-1})$, and $dx/dw = w/(2A)$ turns this into an $O(w_0^{-M})$ error in the smooth root.

At nearby integers the relative forward error in (156) is $O(w_0^{-M-1})$. Moving the smooth root by $h = B w_0^{-M}$ changes its logarithmic value by at least a fixed positive constant times $B w_0^{-M-1}$. Choosing B large enough dominates that error. If x_* is the exact large root of $F_M(x) = y$, then

$$n_- := \lceil x_* - h \rceil - 1, \quad n_+ := \lceil x_* + h \rceil \quad \text{satisfy} \quad X(n_-) < y \leq X(n_+).$$

The same conclusion remains valid if either test integer is farther from the root, since the smooth function is increasing and the displacement is still at most $1 + h$.

The sequence a is globally weakly increasing by (105). For D , use (107) through relative order one, including an $O(t^2)$ remainder. The leading factor $\mathcal{B}(n)t/2$ has ratio $1 + t + O(t^2)$ at consecutive indices; the ratio of its remaining factors is $1 + O(t^2)$, by comparing two bounded remainders directly. Consequently

$$\frac{D(n+1)}{D(n)} = 1 + t + O(t^2) > 1$$

eventually. No error term is differentiated. Also D is eventually positive and unbounded. The finite initial segment is below all sufficiently large y , so the two test integers bracket the first threshold for either X . Absorb the smooth-root error into B_M to obtain (163). $\square$

The interval before taking ceilings shrinks to zero. Its endpoint ceilings are eventually equal or adjacent, but targets arbitrarily near an integer boundary prevent an unconditional prescription $\nu_X(y) = \lceil x_M(y) \rceil$. The theorem is not a certified finite- y numerical error bound. Deficit monotonicity is only eventual: exact counting gives $D(17) = 44$ and $D(18) = 43$.

30.2 Explicit inverse corrections

For the deficit the first two coefficients are

$$v_1^{(D)} = 3 - \frac{A}{3}, \quad v_2^{(D)} = \frac{21}{2} - \frac{11A^2}{18}.$$

Thus a smooth inverse center has expansion

$$x_D(y) = \frac{1}{24} + \frac{w_0^2}{4A} + \frac{3}{2A} - \frac{1}{6} + \frac{1}{w_0} \left(\frac{21}{4A} - \frac{11A}{36} \right) + O(w_0^{-2}), \quad (164)$$

with the deficit core from (157). For a ,

$$v_1^{(a)} = A + 1, \quad v_2^{(a)} = \frac{5A^2}{6} + \frac{5}{2},$$

and its own core from (158) gives

$$x_a(y) = \frac{1}{24} + \frac{w_0^2}{4A} + \frac{1}{2} + \frac{1}{2A} + \frac{1}{w_0} \left(\frac{5A}{12} + \frac{5}{4A} \right) + O(w_0^{-2}). \quad (165)$$

These are expansions of smooth centers in the sense of (162), not integer thresholds.

Proposition 30.2 (Exact two-index shortcut). *Let $\nu_p(y) = \min\{n \geq 0 : p(n) \geq y\}$. For every real $y > 1$,*

$$\nu_p(y) \leq \nu_a(y) \leq \nu_p(y) + 1. \quad (166)$$

Proof. Set $j = \nu_p(y)$. For $n < j$, $a(n) \leq p(n) < y$, while $a(j+1) \geq p(j) \geq y$. Since $y > 1 = p(0) = p(1)$, no zero-index convention interferes. $\square$

Hence evaluating $a(\nu_p(y))$ selects the correct one of the two candidates once the partition threshold is known.

The smooth inverses for p and a have the same $C = A/\sqrt{3}$, $\beta = 2$, and the same Lambert core. For p , $d_1 = -1$ and $d_j = 0$ for $j \geq 2$, so $v_1 = 1$, $v_2 = 5/2$. Subtracting the corresponding centers gives

$$x_a(y) - x_p(y) = \frac{1}{2} + \frac{5A}{12w_0} + O(w_0^{-2}). \quad (167)$$

The half-index shift concerns smooth centers. The exact discrete statement is (166).

Remark 30.3 ([write] The inverses and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instances, and strict increase of a .* For $X = D$ and y beyond the initial segment, $\nu_D(y)$ is the integer staircase of `p0:def:three-inverses`, so `p0:thm:staircase(1)` applies (in the shipped table D increases strictly for $18 \leq n < 1000$, a finite observation). For $X = a$ the write proves that a is strictly increasing for every $n \geq 0$, so ν_a is the staircase with $n_1 = 0$. *Proof.* By (105), $a(n+1) \geq p(n) = a(n) + D(n)$, so it suffices that $D(n) \geq 1$. For odd $n \geq 5$ the partition $(n-2, 2)$ is counted by $D(n)$ ($n-2$ is odd); for even $n \geq 8$ so is $(n-5, 3, 2)$ ($n-5 \geq 3$ is odd and the minimum is 2). Hence $a(n+1) > a(n)$ for $n = 5$ and all $n \geq 7$, and the values $a(0), \dots, a(8) = 0, 1, 2, 3, 5, 6, 11, 12, 20$ settle $n \leq 7$. (This strengthens the weak monotonicity stated after (105); note $D(6) = 0$.)

- (b) *Instance.* The Lambert root (159) is `p0:thm:lambert-core(3)`, case $b_{\text{vol}} < 0$, with $X_{\text{vol}} = w_0$, $a_{\text{vol}} = 1$, $b_{\text{vol}} = -\beta$, $L_{\text{vol}} = \log(y/C)$; the condition $y > C(e/\beta)^\beta$ is the fold condition.
- (c) *Formal instance.* The equation (160) is the formal-exactness identity of `p0:thm:lambert-centered(4)` with $a_{\text{vol}} = 1$, $b_{\text{vol}} = -\beta$, $Q_{\text{vol}}(\tau) = \log(1 + \sum_j d_j \tau^j)$; it is Part I's (54) with $d = \beta$ and $d_j = (2A)^j c_j$, so v_m is Part I's u_m , and (161) is (55)–(57) in these variables (checked symbolically). The smooth root in the proof of Theorem 30.1 is `p0:thm:lambert-centered(5)` in w (the truncated F_M is exactly of exponential–power type in w), followed by $x = \frac{1}{24} + w^2/(4A)$; its residual-over-slope step is `p0:thm:backward-error`.
- (d) *Analogue.* Theorem 30.1 is, as stated and proved, an analogue of `p0:thm:staircase(2)`.
- (e) *Not shown to be an instance.* Proposition 30.2 compares the staircases of p and a directly through the injection behind (105); it involves no smooth model, and the write knows no theorem of the volume of which it is an instance.

No novelty is claimed for any of these inversions.

31 A rational certificate at a finite index

Theorem 26.1 becomes a rational certificate for rational r and $K \in \{2, 3, 4\}$. For these K , the nonzero gaps $1 - \cos(2\pi h/k)$ are rational:

$$k = 2 : (2), \quad k = 3 : (3/2, 3/2), \quad k = 4 : (1, 2, 1).$$

More generally, rational lower bounds for the gaps would suffice. Here no such approximation is needed.

Choose integers $J \geq 1$ and $b \geq 1$, set $S = 10^b$, and define integer upward roundings

$$V_0 = S, \quad V_j = \left\lceil \frac{V_{j-1}}{1 - r^j} \right\rceil \quad (1 \leq j \leq J).$$

Then $V_J/S \geq \prod_{j=1}^J (1 - r^j)^{-1}$. The omitted tail satisfies

$$\begin{aligned} \log \prod_{j>J} (1 - r^j)^{-1} &= \sum_{j>J} \sum_{h \geq 1} \frac{r^{jh}}{h} \\ &\leq \sum_{j>J} \frac{r^j}{1 - r^j} \leq \delta_J := \frac{r^{J+1}}{(1 - r)(1 - r^{J+1})}. \end{aligned}$$

If $\delta_J < 1$, then $e^{\delta_J} \leq (1 - \delta_J)^{-1}$, whence

$$P(r) \leq \bar{P}(r) := \frac{1}{S} \left\lceil \frac{V_J}{1 - \delta_J} \right\rceil. \quad (168)$$

For rational $x \geq 0$ and an integer $H \geq 0$,

$$e^{-x} \leq \left(\sum_{h=0}^H \frac{x^h}{h!} \right)^{-1}. \quad (169)$$

Every operation in (168) and (169) is exact rational arithmetic or an integer ceiling. Substituting these upper bounds for the positive factors in (125) yields a rational upper bound $\bar{\mathcal{E}}$. Set $E = \lceil \bar{\mathcal{E}} \rceil$. Because $D(n)$ is an integer, the certified enclosure is

$$\lceil s_K(n) - E \rceil \leq D(n) \leq \lfloor s_K(n) + u_K(n) + E \rfloor. \quad (170)$$

This proves the certificate procedure, independently of any numerical exponential evaluation.

For $n = 1000$, $K = 3$, $r = 24/25$, the supplied program uses $b = 100$, chooses the first J with $\delta_J < 10^{-30}$, and uses $H = 140$. It obtains $J = 1771$ and the following exact values:

$$\begin{aligned} s_3(1000) &= \frac{1420175722807416251109237414763}{3}, \\ u_3(1000) &= 7689741909994010189877860653, \\ E &= 666467645896325327332069792. \end{aligned}$$

Therefore

$$\begin{aligned} 472725439956575758375747068463 &\leq D(1000) \\ &\leq 481748117158362419220289068699. \end{aligned} \tag{171}$$

Independent exact counting gives

$$D(1000) = 479221051806021018167049012329.$$

The width reflects the elementary Fourier bound; it is not an estimate of the unknown asymptotic remainder. The certification concerns (171) only, not effective constants in Theorem 30.1.

Corollary 31.1 (Certified threshold decision). *Let $y > 1$, $m = \nu_p(y)$, and choose $K \geq 2$ with $m > K(K + 1)/2$. Suppose E is any rigorously evaluated upper bound for $\mathcal{E}_K(m, r)$. Then*

$$\begin{aligned} p(m) - s_K(m) - u_K(m) - E \geq y &\implies \nu_a(y) = m, \\ p(m) - s_K(m) + E < y &\implies \nu_a(y) = m + 1. \end{aligned}$$

If neither comparison settles the question, evaluate $a(m)$ exactly and test $a(m) \geq y$. For a rational target and rational enclosures, every comparison can be performed with exact rational arithmetic.

Proof. Subtract the two sides of (126) from $p(m)$ to obtain a lower and an upper bound for $a(m)$. Apply Proposition 30.2. Equality belongs to the first case because the threshold convention is weak; the second case requires a strict inequality. $\square$

This is an effective decision procedure once m is known. It does not require effective constants in the asymptotic inverse theorem.

32 Reproducible exact computations

The accompanying archive contains the article source, PDF, standard-library Python implementation, deterministic generated tables, and a manifest. Running the reproduction script computes all outputs anew. Saved tables or receipts are never used as arithmetic inputs. Integer and rational checks use explicit exceptions, so optimization with `python -O` cannot disable them. The program validates mathematical inputs before computation; in particular booleans are not accepted as integers.

32.1 Two counting constructions

The first exact algorithm implements (104). For each minimum k , it incrementally builds the bounded-part product $\prod_{j=k}^M (1 - q^j)^{-1}$ by the usual unbounded coin-change recurrence. At maxima M divisible by k , it adds the shifted coefficients for the two removed endpoints. It also adds the

singleton terms separately. The unrestricted $p(n)$ table uses an independent standard partition recurrence.

The second algorithm uses conjugation. If μ is conjugate to λ , then

$$\max \lambda = \# \text{parts of } \mu, \quad \min \lambda = \text{multiplicity of the largest part of } \mu.$$

Before adding a possible largest part M , a two-dimensional table records partitions into parts $< M$ by total size s and length ℓ . Adding exactly k copies of M contributes size kM and length $k + \ell$; it is admissible exactly when $\ell \equiv 0 \pmod{k}$. Summing this condition counts $a(n)$ independently of the minimum–maximum product algorithm. Direct enumeration at small sizes gives a third check. These finite checks validate implementations and examples; the theorems rest on the analytic proofs above.

32.2 Independent coefficient checks

The radial coefficients are computed from the truncated exponential-product series in (134). An independent route first constructs the finite q -polynomial $S_{L+1}(q) = \sum_j h_j q^j$ and uses the exact moment formula

$$g_\ell = \frac{1}{\ell!} \sum_j h_j (-j)^\ell \quad (0 \leq \ell \leq L).$$

The terminating derivative polynomials are checked both by the factorial expression (129) and the differential recurrence (131). The algorithm (136) then generates correction coefficients as rational polynomials in A^{-1} . Triangular formal reversion is checked by substituting its coefficients back into (160) with exact rational arithmetic. These residual tests supplement the symbolic proof of the general triangular construction; a finite collection of substitutions alone is not a proof of a symbolic identity for every parameter.

The Fourier identity is checked in the exact group algebra $\mathbb{Q}[z]/(z^m - 1)$, so no approximate complex roots occur. The rational certificate uses only the integer/rational operations proved in Section 31. Separate rational endpoint checks certify the elementary π and e bounds, the $n \geq 90$ onset, and the cutoff inequalities in Section 29; the analytic monotonicity proof extends the endpoint checks throughout their stated ranges. The archive documents check ranges and commands, and includes a deterministic receipt that records their outcomes.

32.3 Scope of reproducibility

The default run computes p, a, D through 1000, with the more expensive independent conjugate comparison through 220. A larger exact count can be requested explicitly. The archive rebuild command regenerates the data, compiles the PDF until consecutive passes have identical bytes, and creates a new archive in a fresh output directory without overwriting its input bundle. A normal run and an optimized run should produce identical semantic receipts and tables. The build uses a fixed timestamp and strips variable PDF metadata to make the packaged result predictable.

No floating-point diagnostic is used to certify a theorem, a remainder constant, a finite- y inverse, or the rational enclosure. The source files contain no private working notes, review reports, or copied source papers. The bibliography supplies public primary-source links instead.

33 Limitations and conclusions

The leading equivalent for $a(n)$ follows immediately from the sandwich (105). The refinement is that the missing partitions have relative mass $\pi/(2\sqrt{6n})$ to first order, with an exact algorithm for

every fixed correction order. The residue obstruction becomes exponentially small at each fixed minimum, and the remaining minimum tail can be made smaller than any prescribed algebraic order by fixing a sufficiently deep truncation.

The quantifiers are essential. The series are asymptotic formal expansions, with no convergence assertion. The fixed-order expansion is not asserted uniform in increasing minimum, modulus, or order. The separate stretched-exponential comparison uses a growing auxiliary cutoff only through exact uniform inequalities. It is not an actual-error asymptotic and does not prove an $\exp(-c\sqrt{n})$ rate. There is no global monotonicity assertion for D . The smooth inverse bracket has ineffective constants and onset, and does not license blind rounding at all targets. The exact two-index shortcut for a and the rational finite- n certificate are separate effective statements.

The source comparison establishes close prior frameworks and clearly identifies two access limitations. It does not establish first priority for the specialization, coefficients, or combination of results. The outcome is a fully specified application with proofs, exact coefficient generation, and reproducible arithmetic checks.

Several further questions are natural:

- Make the inverse error constants and their onset explicit, beyond the separate finite- n certification method.
- Sharpen local twist bounds and fixed-minimum rates using the finer established twisted-product asymptotics, with explicit treatment of their angular scope.
- Study other minimum weights, first comparing each proposed specialization with existing arbitrary-weight sum-of-tails identities and checking its tail control.

[write] **7 October 2026.** Under Vladimir’s standing rule of 4 October 2026, the write found no claim of this Part that is unproved as stated and none that is wrong, so nothing was moved to these questions or refuted. Remark 30.3(a) adds a proved fact (strict increase of a for every n), not an answer to any of the questions above.

A Tables regenerated by the companion program

The tables in this appendix are included from files written by the exact reproduction program during the build. The printed expressions in the main text are also tested by that program.

A.1 Exact sample counts

n	$p(n)$	$a(n)$	$D(n)$
10	42	37	5
20	627	556	71
50	204226	187251	16975
100	190569292	179039597	11529695
200	3972999029388	3800030996209	172968033179

Quantity	Exact value
$p(500)$	2300165032574323995027
$a(500)$	2235865358882256076332
$D(500)$	64299673692067918695
$p(1000)$	24061467864032622473692149727991
$a(1000)$	23582246812226601455525100715662
$D(1000)$	479221051806021018167049012329

A.2 Radial coefficients

ℓ	g_ℓ	ℓ	g_ℓ
1	$\frac{1}{2}$	7	$\frac{21841}{10080}$
2	$\frac{1}{12}$	8	$\frac{10023809}{1209600}$
3	$\frac{1}{12}$	9	$\frac{26596837}{725760}$
4	$\frac{89}{720}$	10	$\frac{44079183131}{239500800}$
5	$\frac{61}{240}$	11	$\frac{82627721881}{79833600}$
6	$\frac{20287}{30240}$	12	$\frac{8426813263570739}{1307674368000}$

A.3 Finite enclosure and check receipt

Quantity	Exact value
Lower bound	472725439956575758375747068463
$D(1000)$	479221051806021018167049012329
Upper bound	481748117158362419220289068699
Product factors	1771
Exponential degree H	140

Exact check	Range or count
Euler recurrence versus partition knapsack	$0 \leq n \leq 1000$
Original extrema versus conjugate statistic	$0 \leq n \leq 220$
Direct partition enumeration	$1 \leq n \leq 35$
Fourier reduction in the integral group ring	49 cases, $0 \leq n \leq 70$
Radial product versus finite moments	degree 20
Derivative recurrence versus factorial formula	$Q_0, \dots, Q_{20}$
Independent forward-coefficient generators	$c_0, \dots, c_{12}$
Rational inverse specializations, both statistics	40 cases, degree 12
Exact threshold breakpoint tests	864
Certified inverse endpoint tests	8
Uniform error bound, $n \geq 90$	19 rational endpoint certificates
Uniform certificate invalid-input probes	18
Rejected invalid-input probes	22

[write] **The merged bibliography, 7 October 2026.** The three delivered bibliographies are merged, and every detail of the delivered entries is kept; an entry cited by more than one Part is printed once, with each Part’s own note on what it inspected. Bracketed Part numbers mark entries cited by one Part only. Renamed keys: Part II’s `oeis` and `dlmf` are `oeisA118096` and `dlmfBessel`; Part III’s `oeis` is `oeisA117086`, and its `nr` is Part I’s `ngorhoades`.

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